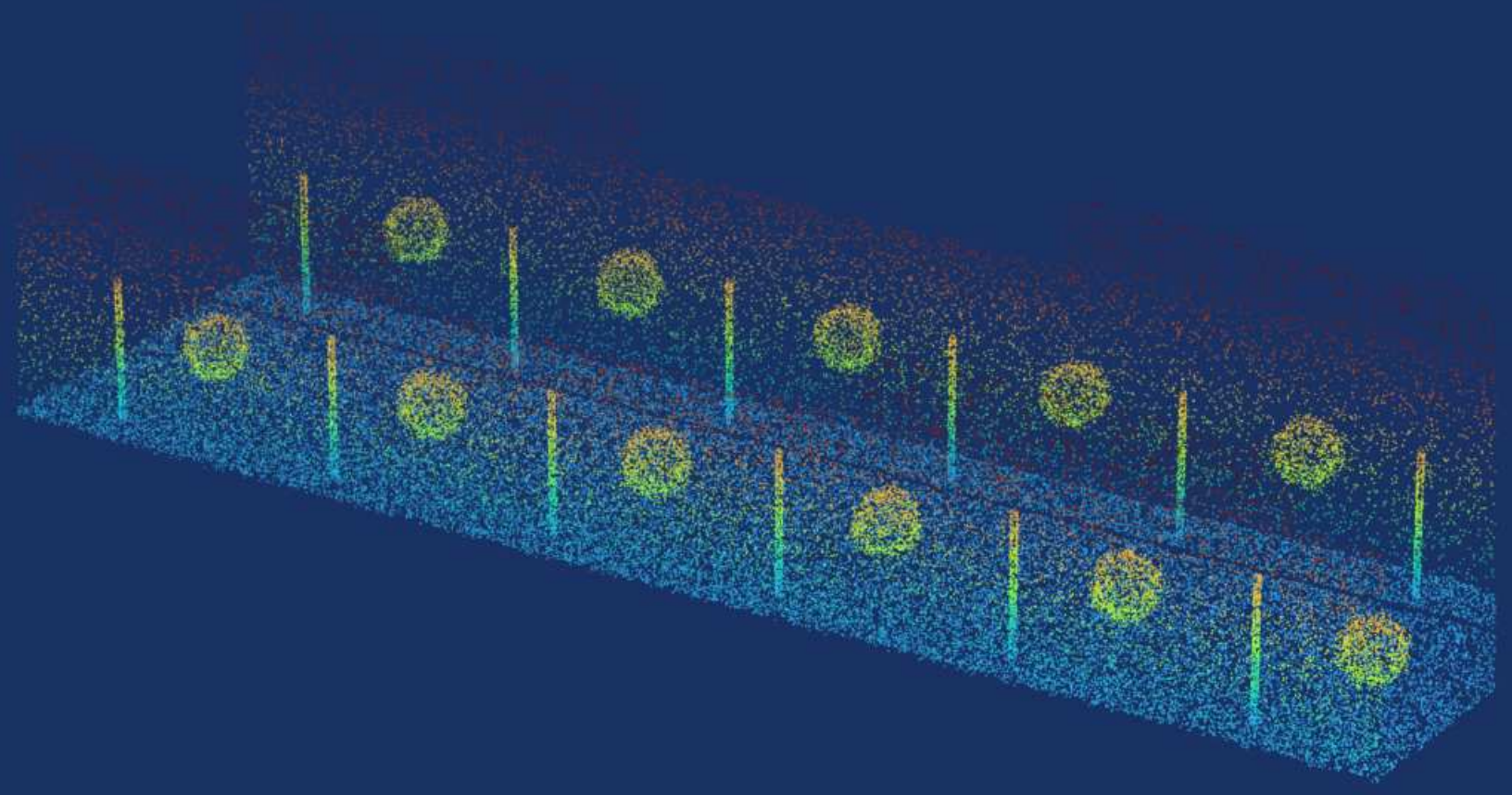
LiDAR from the Ground Up

Surveying Concepts Edition

D.S. Banks, PSM

The core ideas of mobile LiDAR in plain language,
with field tips for surveyors.



21 modules • concepts, field tips and illustrations

Course Text R1

About This Edition

This is the Surveying Concepts edition (R1) of *LiDAR from the Ground Up*. It keeps the plain-language track of the full course: every green **Simple Approach** box, every **Big Idea**, every **Surveyor's Field Tip** and **Warning**, and the illustrations, in the same order and with the same module numbers as the full text.

The engineering track — equations, worked examples, exercises, reference tables, labs, the sample SOP and the appendices — is in the full course text (R0).

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MODULE 1

What Is LiDAR?

SIMPLE APPROACH

Close your eyes and clap your hands in a big gym. You hear an echo. If the wall is far away, the echo takes longer to come back. If it is close, the echo comes back fast. A bat flies in the dark this way!

A LiDAR does the same thing — but instead of a clap it uses a tiny flash of laser light, and instead of ears it has a super-fast light catcher. It does this hundreds of thousands to millions of times every second, in every direction, so it can "see" the shape of everything around it: roads, trees, buildings, people. Each echo becomes one dot. Millions of dots together make a 3-D picture called a **point cloud**.

Now imagine doing the clapping trick while riding in a car. To draw the dots in the right place, you also have to know exactly where the car was, and which way you were facing, every time you clapped. That is the hard part, and it is what this whole course is about.

1.1 The name

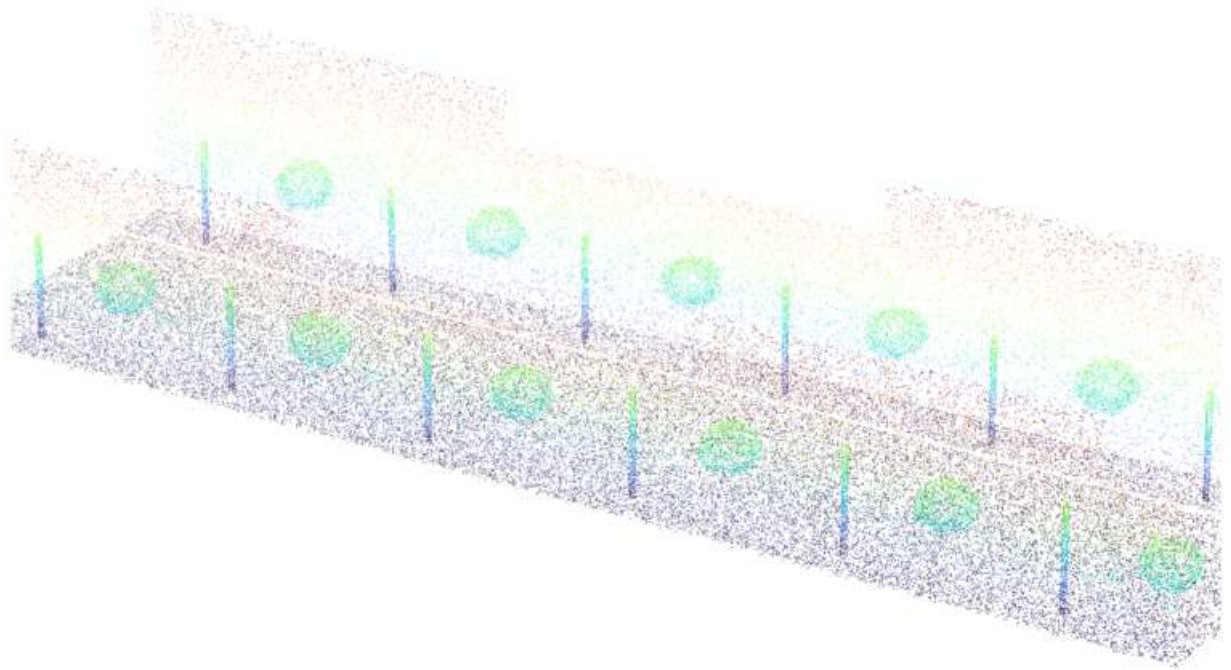


Figure 1.1: A synthetic mobile-LiDAR point cloud of a street, colored by height: road, curbs, poles, trees and building faces. Every dot is one laser echo.

1.2 The four families of LiDAR

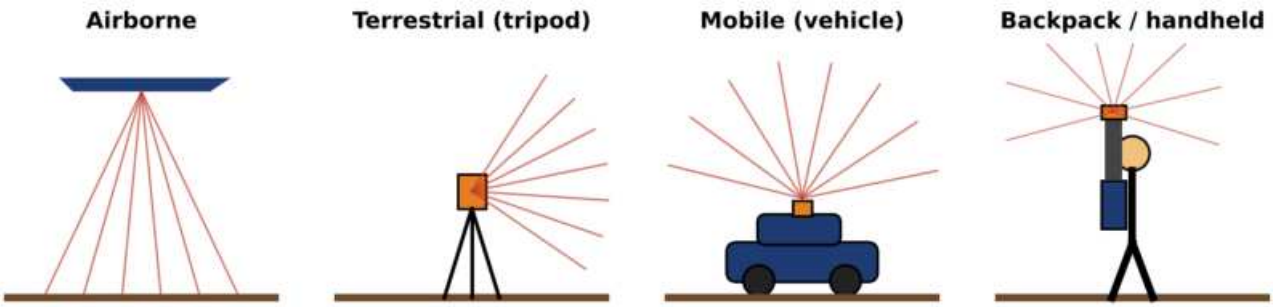


Figure 1.2: The same measuring idea mounted four ways.

1.3 The one big problem: the instrument is moving

BIG IDEA

A laser can only tell you *"that wall is 25.000 m away in that direction from me."* If you do not know exactly where "me" is — and which way "me" was pointing — at the *exact instant* of each pulse, the measurement is useless. Most of this course is about knowing "me": the sensor's **pose**.



Figure 1.3: Left: the same scans placed with a bad trajectory form the famous "bird's nest." Right: with a correct high-rate trajectory every scan stacks into a crisp map.

1.4 How small errors become big errors

BIG IDEA

Angle errors grow with distance and timing errors grow with speed. A system that is excellent at 15 m and walking pace can be unacceptable at 80 m and highway speed. Specifications that leave out range and speed are incomplete.

MODULE 2

The Math Toolkit

2.1 Big and small numbers (scientific notation)

SIMPLE APPROACH

Light is *so* fast that it could go around the Earth about seven times in one second. To write such big numbers — and tiny ones like "one billionth of a second" — scientists use a shortcut: count the zeros and write that count as a little number up high. 10^3 means 1 followed by 3 zeros = 1000. 10^{-3} means "1 divided into 1000 pieces" = 0.001.

2.2 Angles: degrees and radians

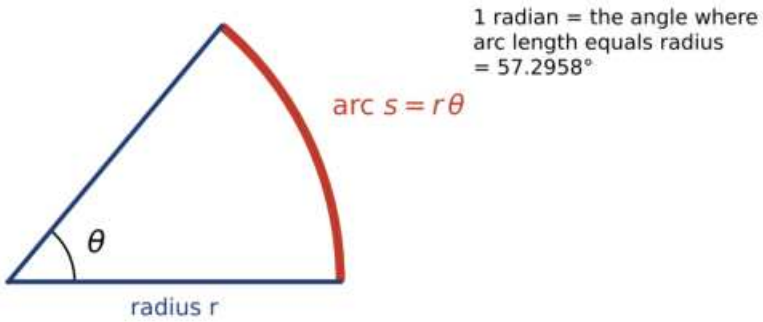


Figure 2.1: A radian is the angle whose arc is exactly one radius long.

SIMPLE APPROACH

Imagine a pizza. If you cut a slice whose crust is exactly as long as the line from the center to the crust, that slice's angle is **one radian**. A whole pizza has a bit more than 6 of those slices (6.28, which is 2π).

2.3 Sine and cosine

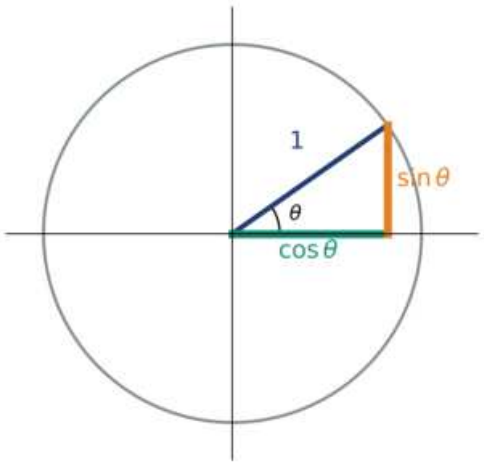


Figure 2.2: On a circle of radius 1, cosine is the “across” part and sine is the “up” part.

SIMPLE APPROACH

Lean a 1-meter stick against a wall. The bottom of the stick is some distance from the wall (that is **cosine**) and the top is some height up the wall (that is **sine**). If the stick leans a lot, cosine is big and sine is small. If it stands almost straight up, sine is big and cosine is small.

2.4 Vectors

SIMPLE APPROACH

A vector is an arrow: it says *how far* and *which way*. “Walk 3 steps east and 1 step north” is a vector. Two walks one after the other add up to one bigger walk.

2.5 Statistics: mean, standard deviation, RMSE



Figure 2.3: Random measurement errors usually follow a bell curve. About 68% fall within one standard deviation σ and 95% within 1.96σ.

SIMPLE APPROACH

Throw ten paper balls at a target. The **mean** tells you where they landed on average (are you aiming too far left?). The **spread** (standard deviation) tells you how scattered they are. A great thrower has a mean right on the bullseye *and* a small spread.

MODULE 3

Measuring Distance with Light

3.1 Time of flight

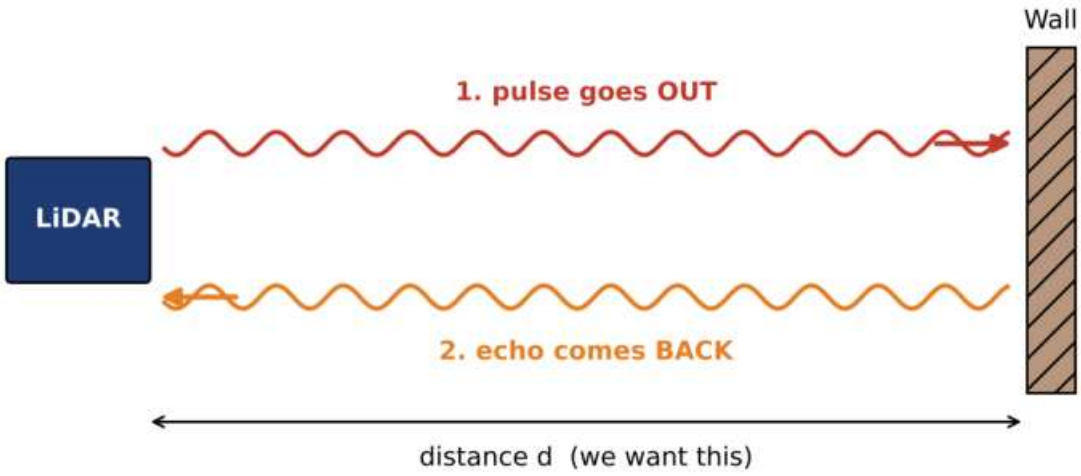


Figure 3.1: A pulse travels out to the wall and back. The clock measures the whole round trip.

SIMPLE APPROACH

Light is the fastest thing there is. In the time it takes you to blink, light could go around the world more than 2 times! The LiDAR starts a stopwatch when it flashes, and stops it when the flash bounces back. Because the light went *to* the wall and *back*, the wall is only half as far as the light traveled. So we cut the answer in half.

3.2 How precise must the clock be?

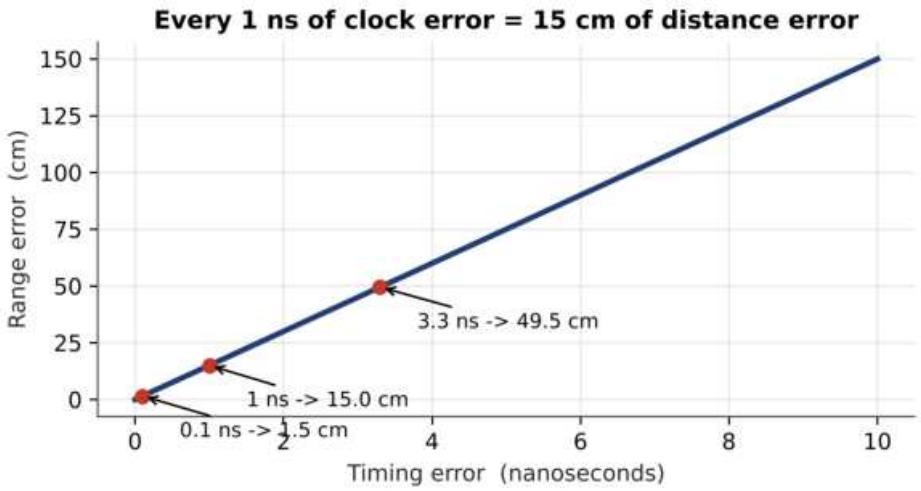


Figure 3.2: Range error grows in a straight line with timing error.

3.3 Pulse rate and range ambiguity

SIMPLE APPROACH

Imagine shouting "Hello!" every second in a giant canyon. If an echo takes 3 seconds, you won't know which "Hello" it belongs to. A LiDAR has the same problem if it fires the next flash before the last echo has come home.

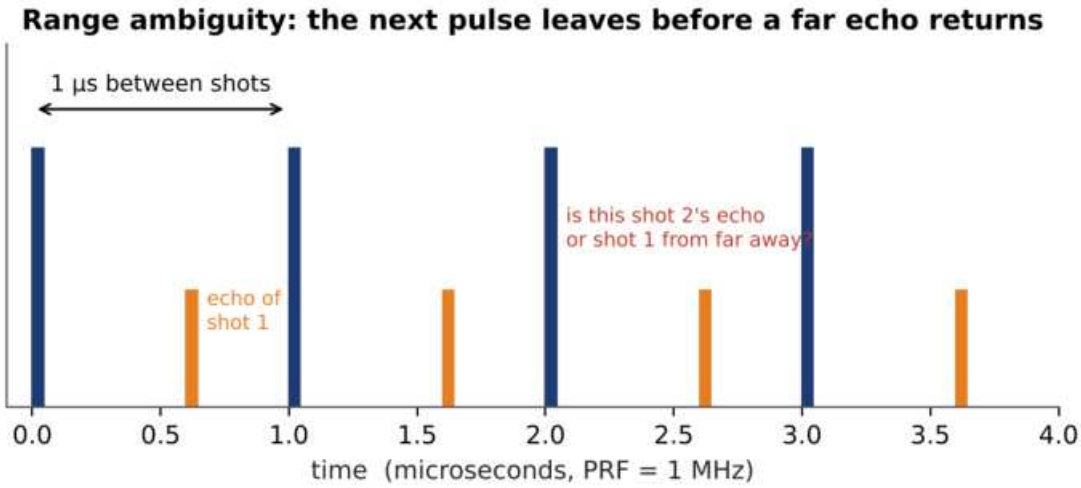


Figure 3.3: If the pulse repetition interval is shorter than the round-trip time, echoes get mixed up.

3.4 Phase-shift (continuous-wave) ranging

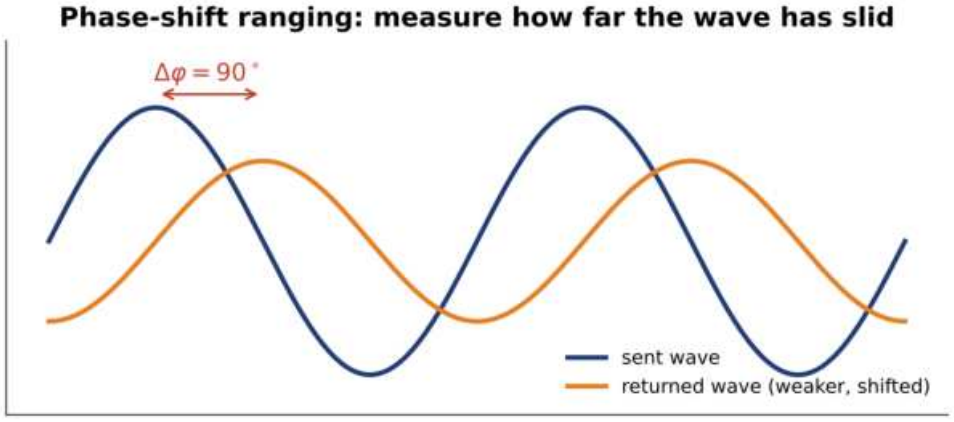


Figure 3.4: The returned wave is shifted by a fraction of a wavelength.

3.5 Multiple returns

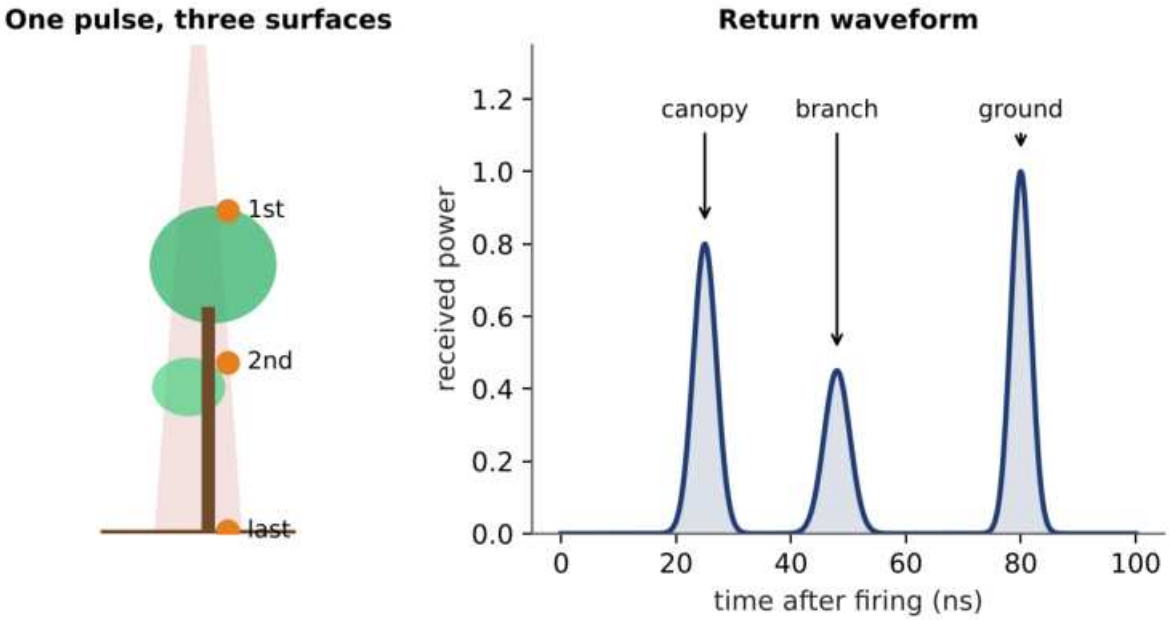


Figure 3.5: One wide pulse can hit leaves, a branch and the ground. The receiver sees several peaks.

SURVEYOR'S FIELD TIP

Edges of poles, curbs and building corners produce "mixed pixels": part of the beam hits the near object and part the far background, and the reported range can land in empty air between them. These show up as ghost points trailing behind edges and are removed in cleaning (Module 18).

MODULE 4

The Laser Beam and the Echo

4.1 The beam spreads out

SIMPLE APPROACH

Shine a flashlight at your hand: you see a small circle. Now shine it at a wall across the room: the circle is much bigger. A laser beam spreads too — just very, very slowly. Far away, the laser "dot" can be as big as a coin, so it can touch the edge of a pole and the wall behind it at the same time.

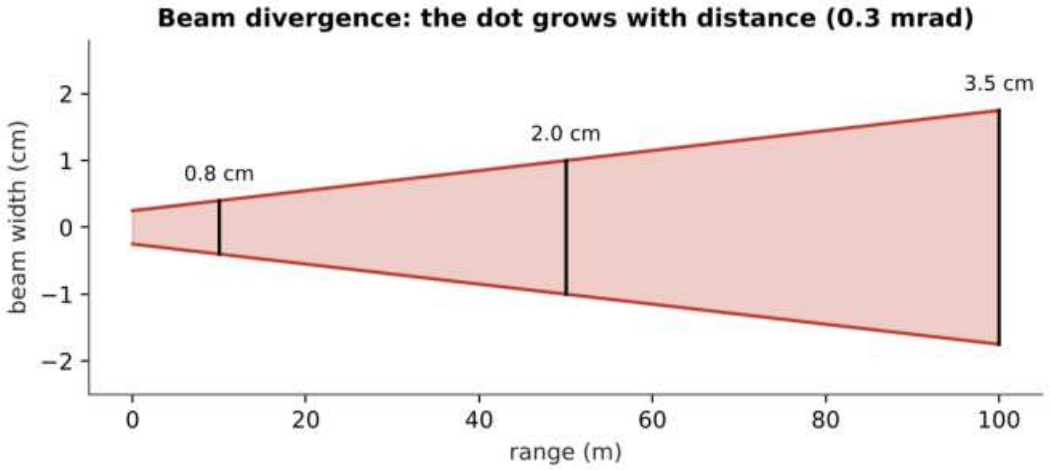


Figure 4.1: The footprint of a beam with 5 mm exit diameter and 0.3 mrad divergence.

4.2 How strong is the echo? The LiDAR range equation

SIMPLE APPROACH

The farther away something is, the dimmer its echo. It gets dim fast: twice as far means only one quarter as bright. Shiny white things (a road sign) send back lots of light; black asphalt or wet surfaces send back very little. And if the light hits a surface at a slant, most of it bounces away like a ball skipping off water.

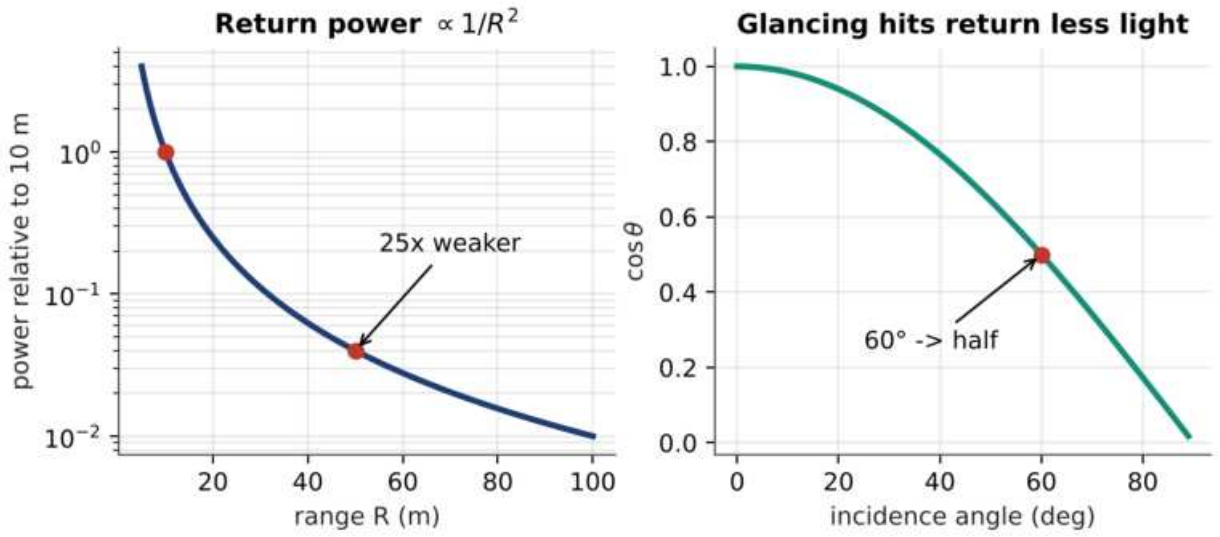


Figure 4.2: Left: the $1/R^2$ law. Right: the $\cos\theta_i$ incidence-angle law.

4.3 What can LiDAR see through?

SIMPLE APPROACH

A laser is light, so it acts like a flashlight beam, not like X-ray vision. It cannot see *through* a wall, a tree trunk, or the ground. It *can* slip through holes — the gaps between leaves, the openings in a fence — and reach whatever is behind them. It passes through clear glass the way sunlight does. And it is stopped almost right away by water.

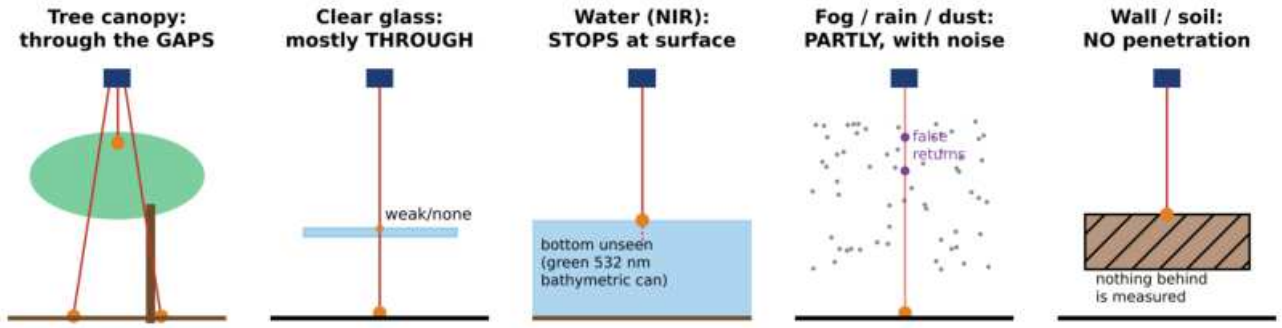


Figure 4.3: The five situations every LiDAR user meets. Orange dots are recorded returns.

BIG IDEA

LiDAR never "penetrates" solid material. It reaches hidden ground only through openings, which is why multiple returns (Module 3) and many viewing angles matter more than laser power.

4.4 Reflectivity of common surfaces

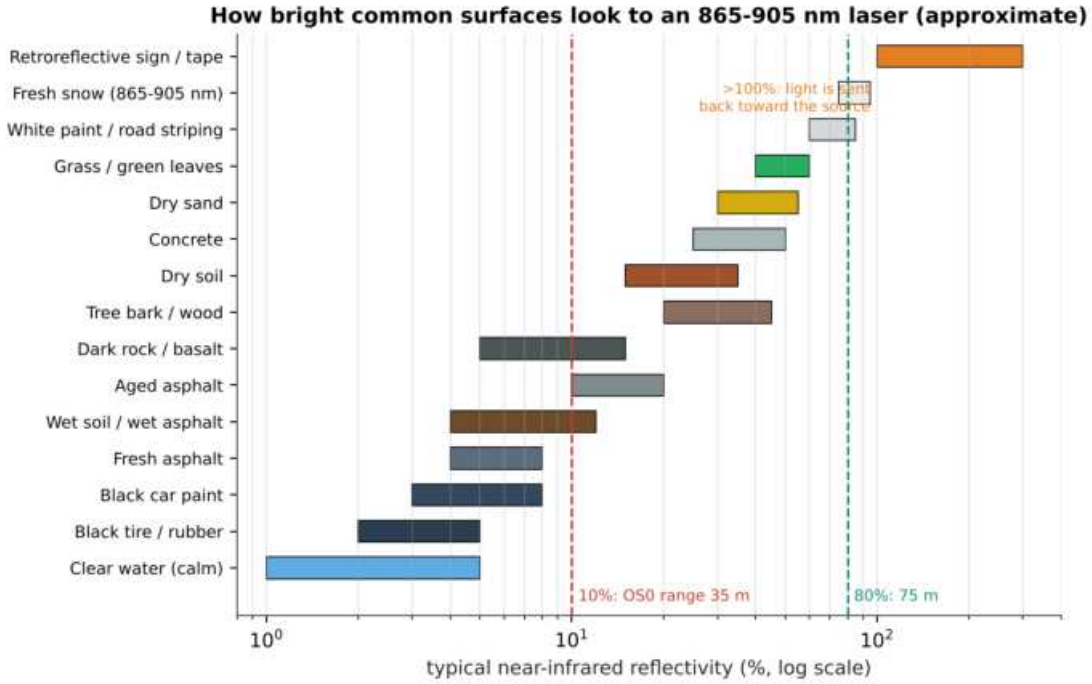


Figure 4.4: Typical reflectivity of common surfaces at the near-infrared wavelengths used by most mobile scanners (865–905 nm). Values vary with moisture, age, roughness, color and incidence angle.

MODULE 5

From Angles and Distance to X, Y, Z

5.1 The idea

SIMPLE APPROACH

To tell a friend where a treasure is hidden, you could say: "Face the big tree. Turn a little to the left. Look a little up. Now walk 25 steps that way." That is a **turn**, a **tilt** and a **distance**. A LiDAR records the same three things for every dot. Then it does some triangle math to change them into "how far forward, how far sideways, how far up" — which we call X, Y and Z.

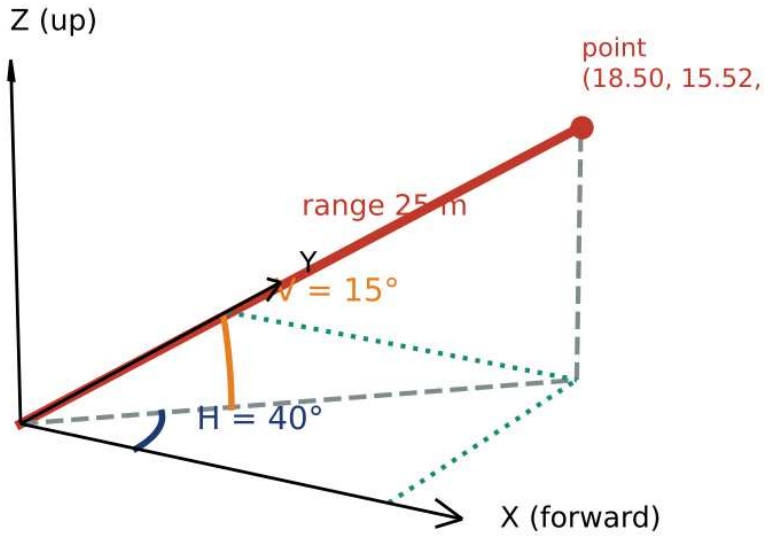


Figure 5.1: A range of 25 m with horizontal angle 40° and vertical angle 15° becomes a point at (18.50, 15.52, 6.47) m.

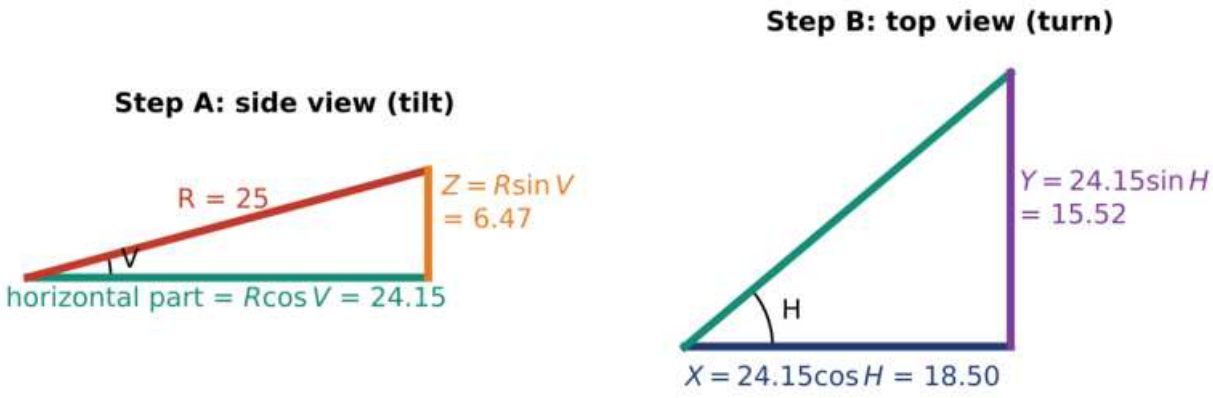


Figure 5.2: The two moves: tilt (side view), then turn (top view).

5.2 Mechanical versus solid-state scanners

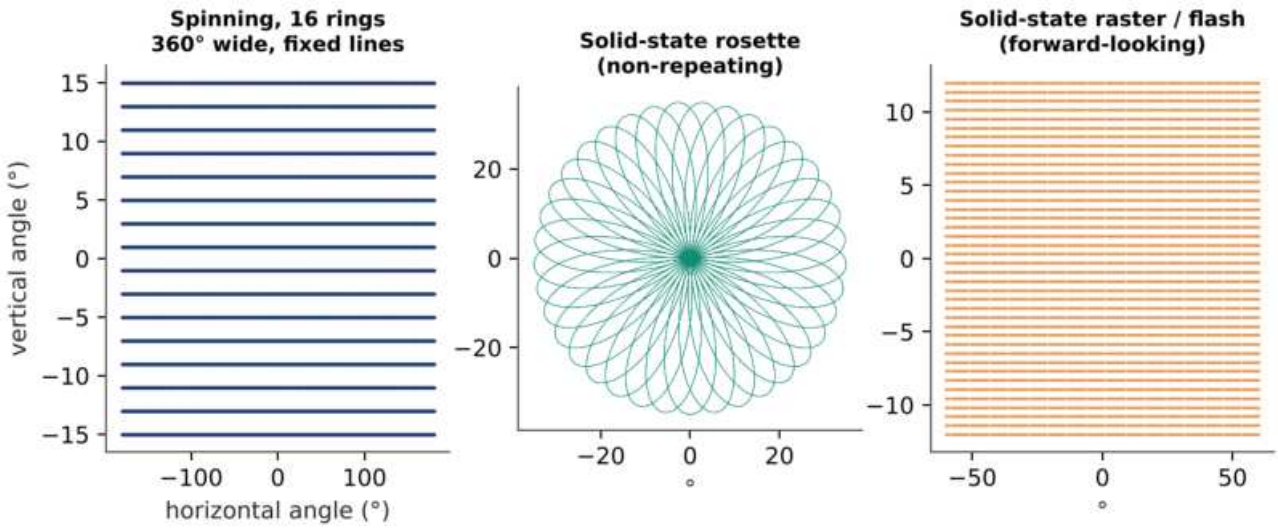


Figure 5.3: How three kinds of scanner sample the world (angles in degrees).

5.3 How dense will my points be?

SIMPLE APPROACH

Imagine painting a road with a paint-roller that only touches the road once every big step. If you walk faster, the stripes get further apart. If you roll faster, they get closer.

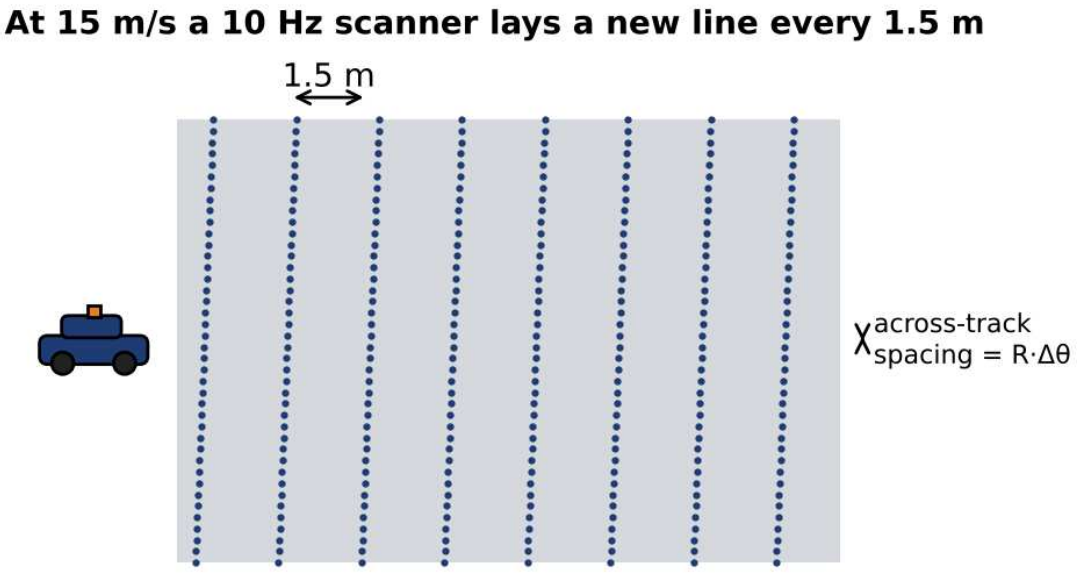


Figure 5.4: Each sweep of the scanner lays a line of points across the road.

SURVEYOR'S FIELD TIP

Mobile mapping heads for survey work are often mounted *tilted* so the spinning plane slices the road at an angle; then several channels hit the same strip of road at different times and along-track gaps close up. Point-density specifications should be checked on the delivered data (points per m^2 on a test patch), not assumed from the brochure.

MODULE 6

Rotation, Pose and Frames

6.1 Pose = where you are + which way you face

SIMPLE APPROACH

Stand in the middle of a room. "Where you are" is your spot on the floor (and how tall you are). "Which way you face" is whether you look at the board or the window, whether you tilt your head, whether you lean. A LiDAR on a car needs to know both of these things, all the time, because the car bumps, turns and leans. Together they are called its **pose**.

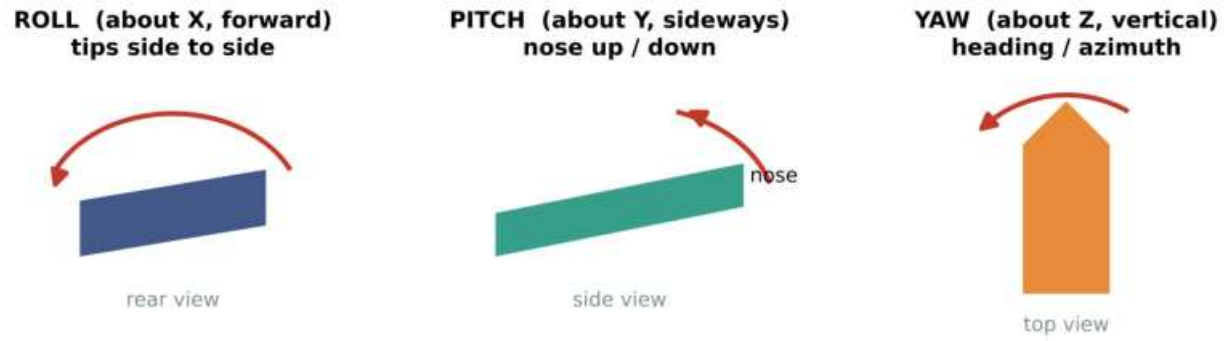


Figure 6.1: Roll, pitch and yaw on a mapping vehicle.

6.2 The right-hand rule: naming X, Y, Z and roll, pitch, yaw

SIMPLE APPROACH

Hold up your **right** hand. Point your **first (index) finger** straight ahead — that is **X**. Point your **middle finger** to the left — that is **Y**. Stick your **thumb** straight up — that is **Z**. **The thumb is always Z.** Another way to say it: curl your fingers from X toward Y, and your thumb points to Z. Now make a thumbs-up: the way your curled fingers wrap around your thumb is the "plus" direction for turning around that thumb. Point your thumb along X and your fingers show plus **roll**; along Y, plus **pitch**; along Z (straight up), plus **yaw**.

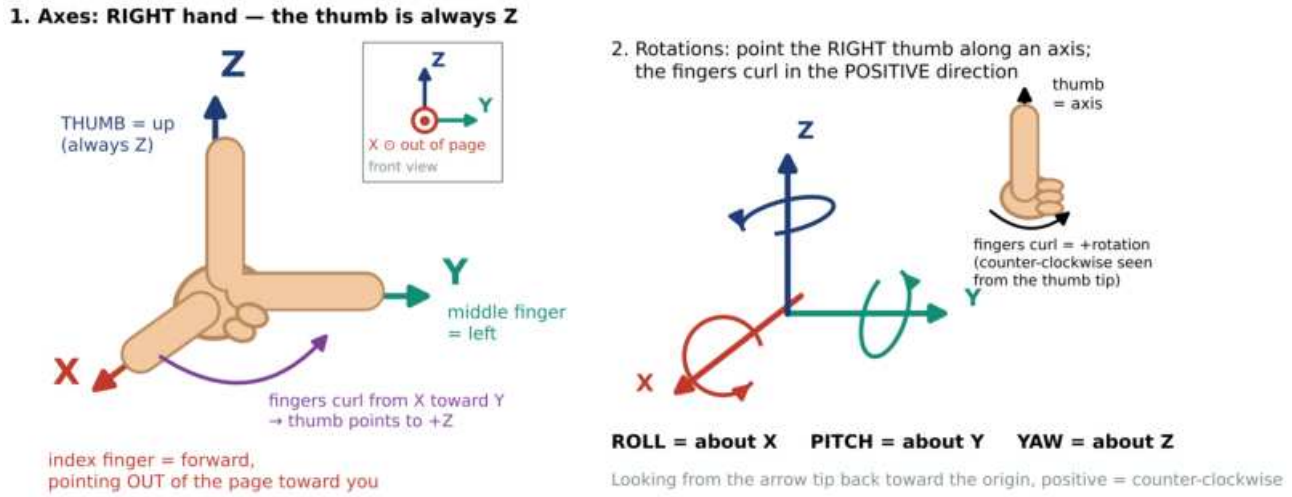


Figure 6.2: The right-hand rule. Left: index finger = X, middle finger = Y, and the thumb is always Z; equivalently, the fingers curl from X toward Y and the thumb points to Z. Right: with the thumb along an axis, the curled fingers show the positive rotation about it — roll about X, pitch about Y, yaw about Z.

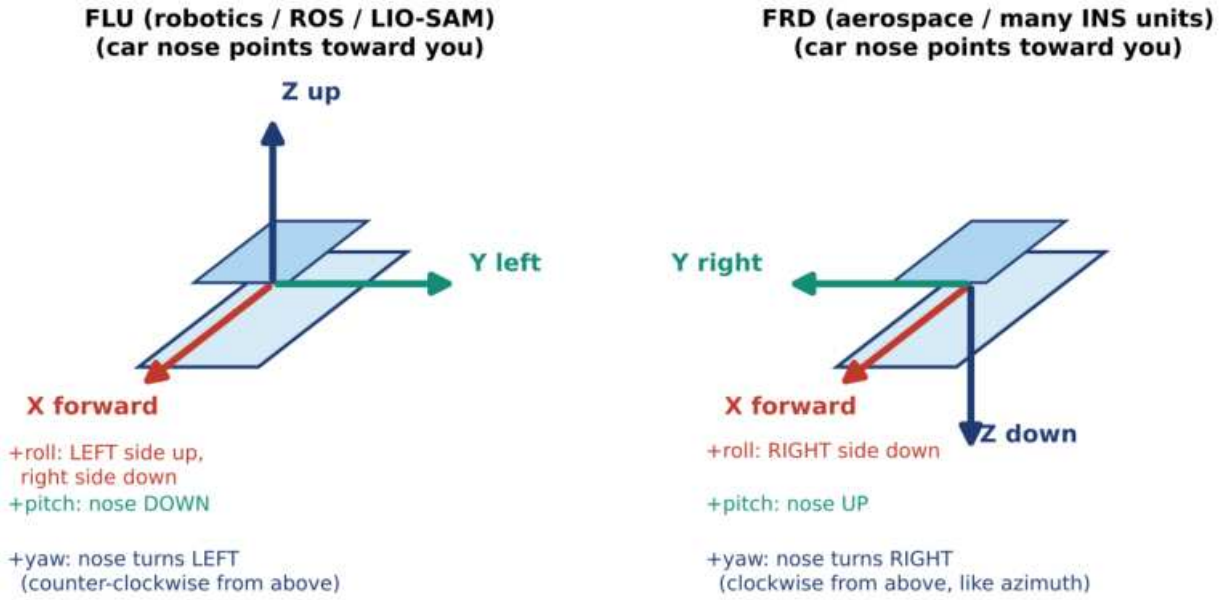


Figure 6.3: The same vehicle in the two common right-handed body frames.

SURVEYOR'S FIELD TIP

When setting up a rig, stand behind the vehicle, hold up your right hand in the way the manual defines the axes, and *physically* roll, pitch and yaw the IMU a few degrees while watching the live output. Each angle should increase in the direction your fingers curl. Write the verified convention in the calibration record (Appendix E).

6.3 Rotating a point in 2-D

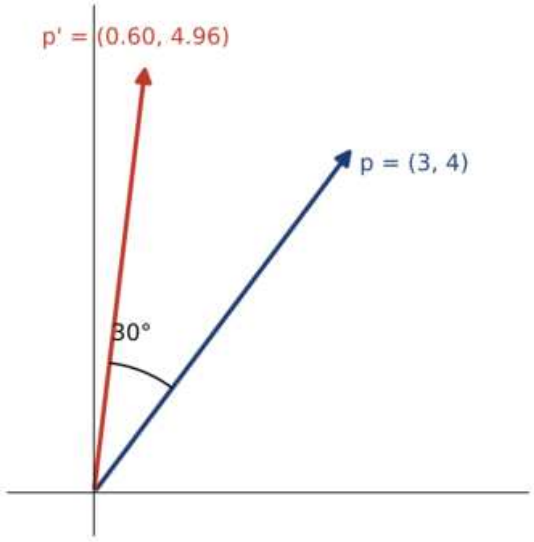


Figure 6.4: Rotating $\mathbf{p} = (3, 4)$ by 30° counter-clockwise.

6.4 Rotations in 3-D

SURVEYOR'S FIELD TIP

Yaw is not azimuth. Surveyors measure azimuth clockwise from grid north. ROS/ENU systems measure yaw counter-clockwise from east. Convert with $\psi = 90^\circ - \alpha_{\text{az}}$. A vehicle heading due north has $\alpha_{\text{az}} = 0^\circ$ but $\psi = 90^\circ$. Grid north also differs from true north by the *convergence angle* of your map projection — another place a heading can quietly go wrong.

6.5 Why tiny angle errors matter

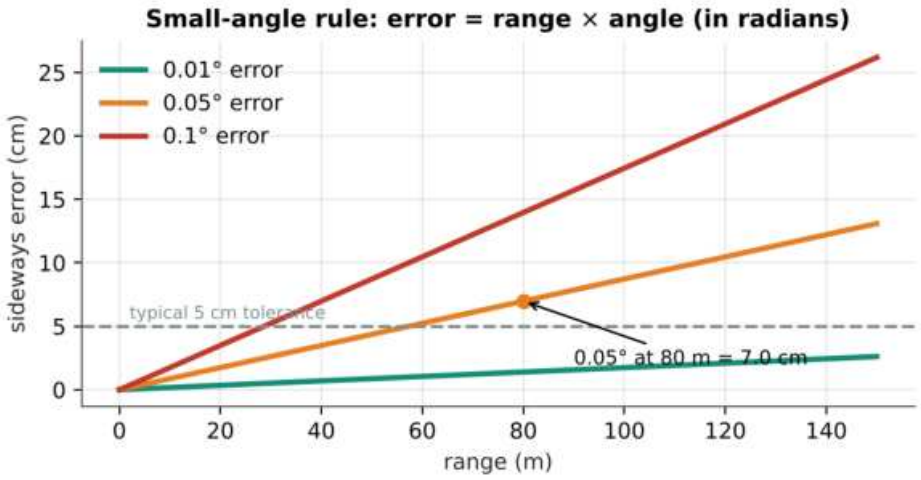


Figure 6.5: Lateral error versus range for three residual angle errors.

MODULE 7

The IMU: Feeling Motion

7.1 What is inside an IMU?

SIMPLE APPROACH

When a car speeds up, you get pushed back into your seat. When it turns, you lean to the side. Your body *feels* the motion even with your eyes closed. An IMU is a little box that feels motion the same way. It has three "push-feelers" (accelerometers) — forward, sideways, up — and three "spin-feelers" (gyroscopes). By adding up all those feelings many times a second, the computer can guess where the car went, even inside a tunnel. But like walking with your eyes closed, the guess slowly gets worse.

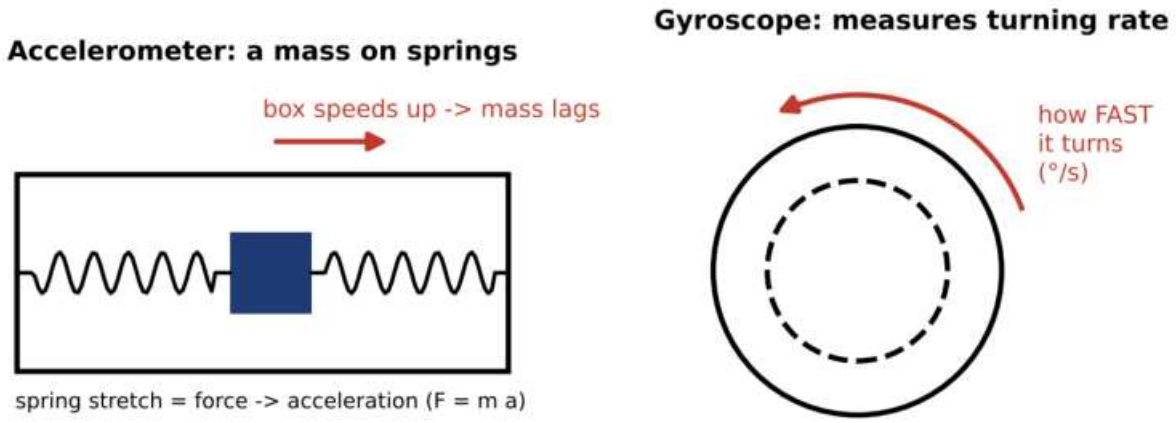


Figure 7.1: An accelerometer is a mass on springs; a gyroscope senses turning rate.

7.2 Strapdown integration (LIO-SAM eqs. 4–6)

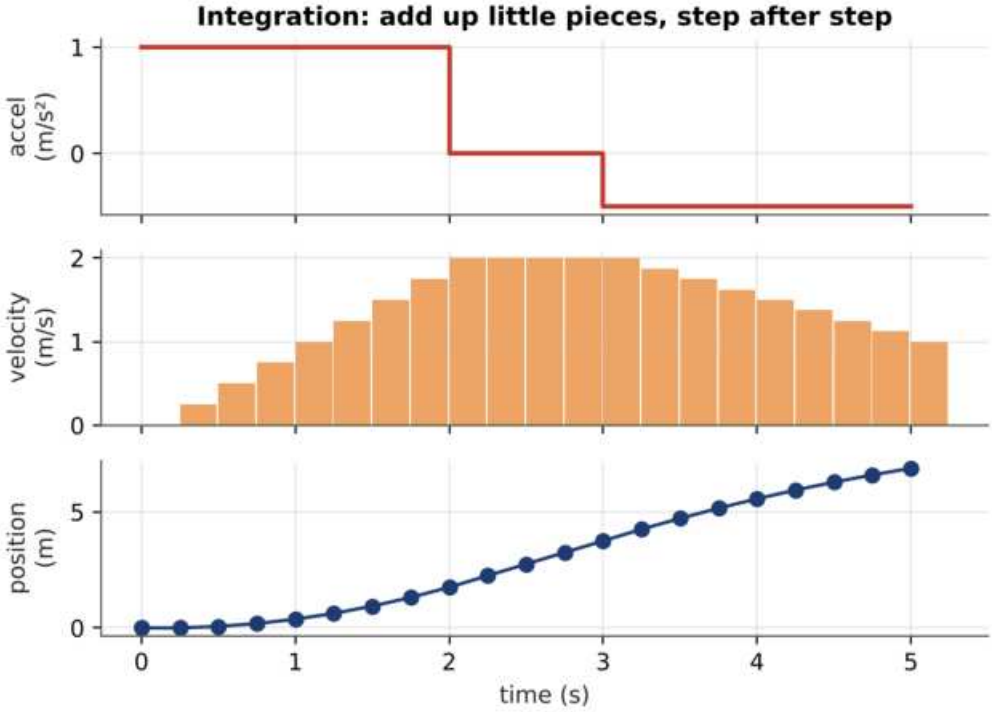


Figure 7.2: Acceleration is summed into velocity; velocity is summed into position.

7.3 Why IMUs drift: biases integrate

SIMPLE APPROACH

Imagine a scale that always reads 1 gram too heavy. Weigh one candy and the mistake is tiny. Weigh a thousand candies one by one and add them up — now you are off by a whole kilogram! An IMU adds up its readings thousands of times, so even a tiny error grows and grows.

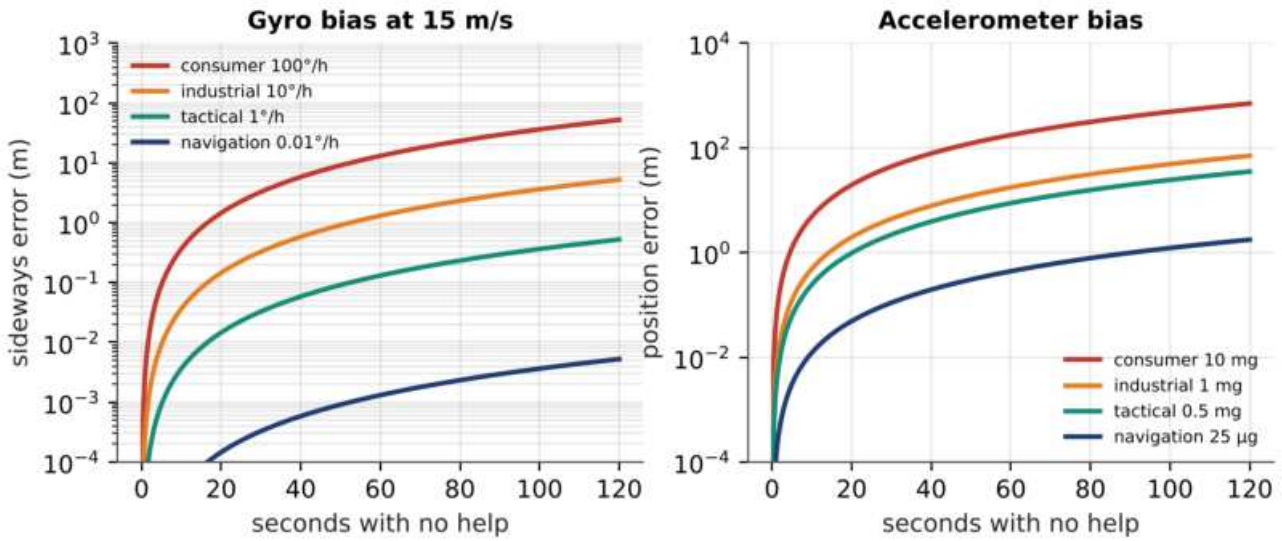


Figure 7.3: Position error versus unaided time for four IMU grades (log scale). Each grade is about ten times better than the one before.

7.4 The hidden killer: tilt error leaks gravity

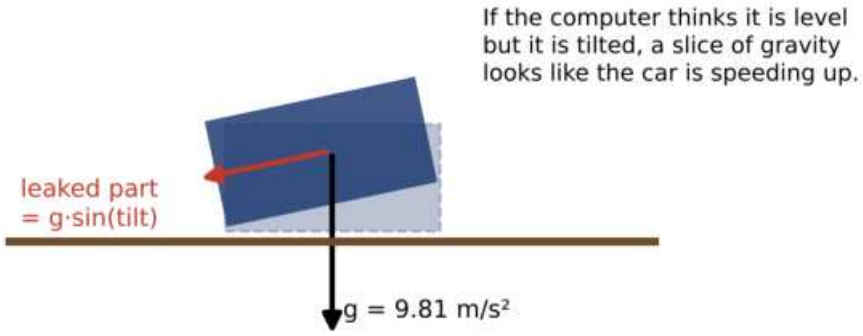


Figure 7.4: A small attitude error makes part of gravity look like acceleration.

7.5 IMU grades

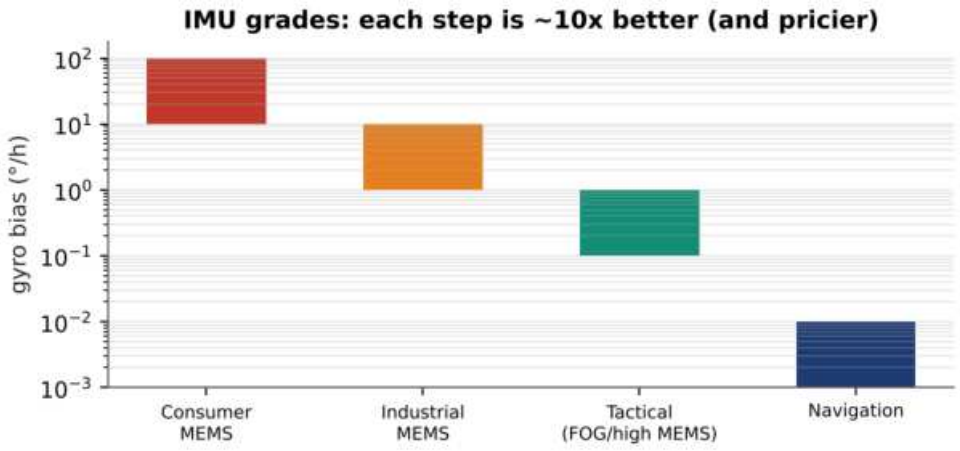


Figure 7.5: Typical gyro bias-instability ranges (order of magnitude).

7.6 IMU preintegration (LIO-SAM eqs. 7–9)

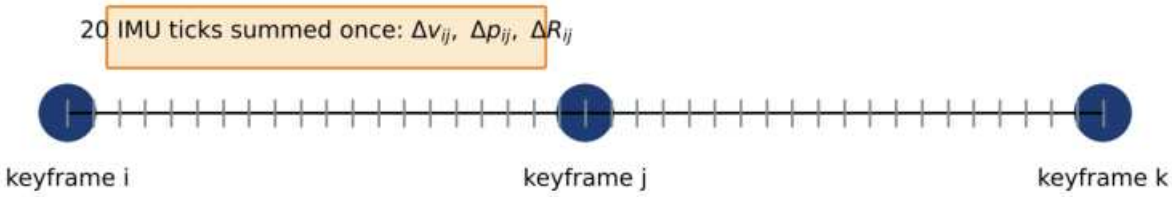


Figure 7.6: Between two keyframes, hundreds of IMU samples are summarized once into a relative motion.

SIMPLE APPROACH

Imagine writing down every step you take on a walk: "step, step, turn a little, step…" Instead of keeping that whole long list, you could write one note: "from the door to the swing: 20 steps forward and a quarter turn." Preintegration is that short note.

MODULE 8

Timing, GNSS and Motion Compensation

8.1 One clock to rule them all

SIMPLE APPROACH

Imagine you and a friend take photos of a race. If your camera's clock is one second slow, your photo of the runner crossing the line will be labeled at the wrong time — and the runner will seem to be in the wrong place. Every sensor on a mapping car must use the *same* clock, set perfectly, so every laser dot gets matched to exactly where the car was at that instant.

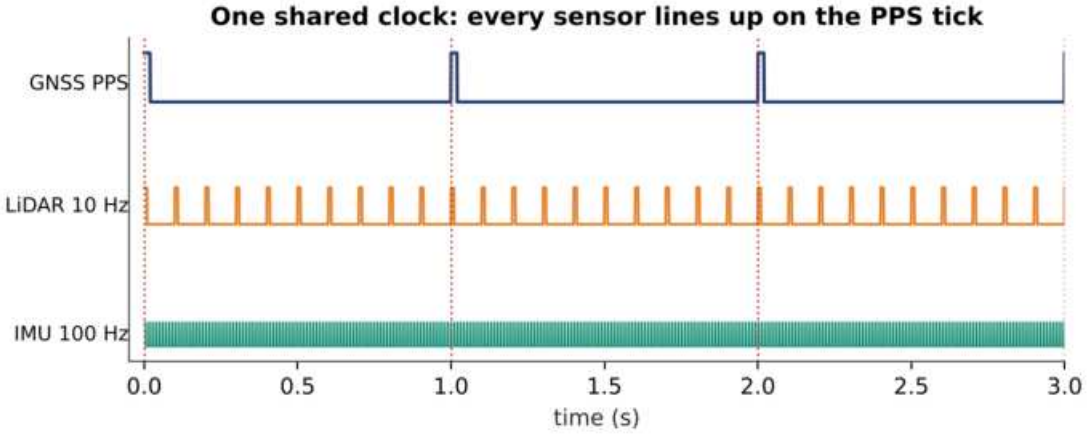


Figure 8.1: GNSS PPS, LiDAR and IMU events all share one time base.

8.2 GNSS and IMU: a perfect partnership

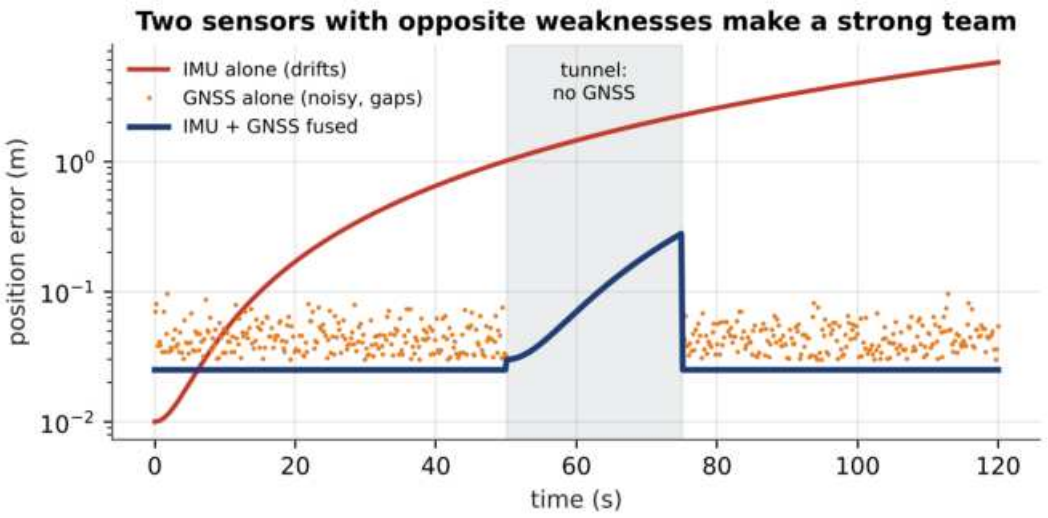


Figure 8.2: The IMU drifts but never stops; GNSS never drifts but can disappear. Together they are far better than either.

8.3 Motion compensation (de-skewing)

SIMPLE APPROACH

Take a photo of a fast car with a slow camera and the car looks stretched and blurry. A spinning LiDAR takes 1/10 of a second to sweep all the way around, and the vehicle moves during that time. If the computer pretends the whole sweep happened at one instant, straight walls come out bent. De-skewing fixes the blur by moving each dot back to where it really was.

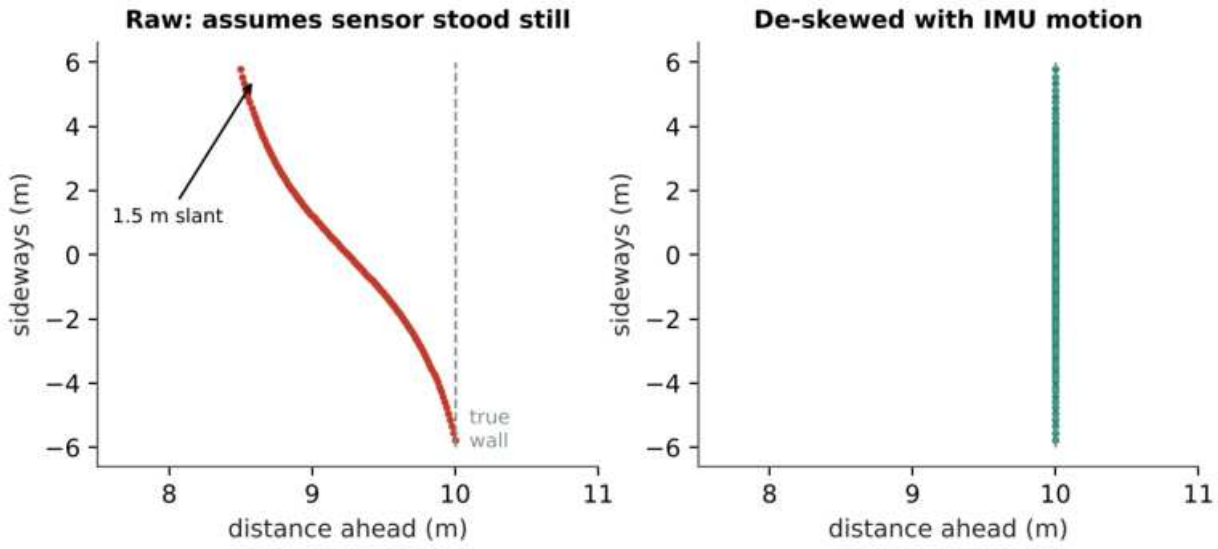


Figure 8.3: A wall scanned while driving toward it at 15 m/s. Left: raw sweep. Right: after de-skewing each point with its own pose.

SURVEYOR'S FIELD TIP

A practical QA check before processing: confirm that (1) the IMU stream is present at its nominal rate with no gaps, (2) every LiDAR point has a per-point time field, (3) LiDAR and IMU timestamps overlap and are in the same time base, and (4) the static lever-arm/boresight transform is loaded. If any of these is missing, stop — no amount of processing will fix it later.

MODULE 9

Lever Arm and Boresight Calibration

9.1 Two boxes on one plate

SIMPLE APPROACH

Put a toy camera on your shoulder and close your eyes. A friend asks, “Where is the camera?” You know where *you* are, but the camera is a little to the side and a little higher — and it might be turned slightly. If you forget that, you will point to the wrong place. The car’s “body” (the IMU) and its laser “eye” are in different spots too. The **lever arm** is how far apart they are. The **boresight** is how much one is twisted compared to the other.

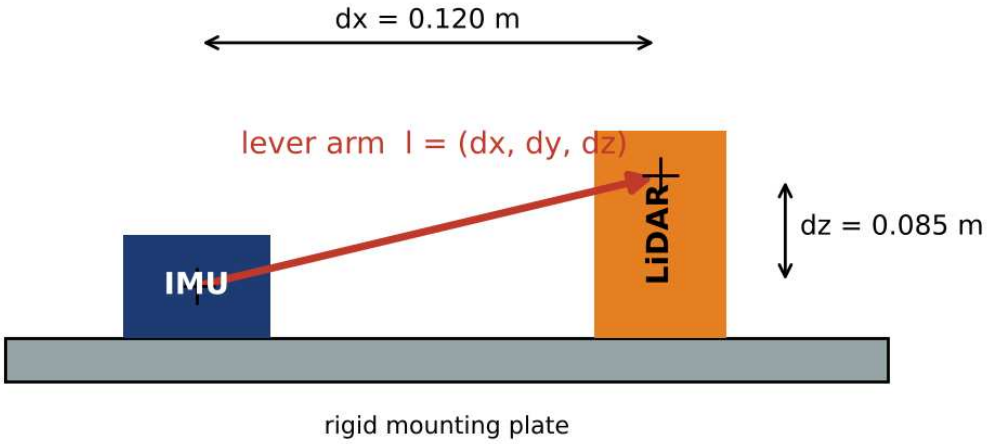


Figure 9.1: The lever arm is the vector from the IMU origin to the LiDAR origin.

9.2 The direct georeferencing equation

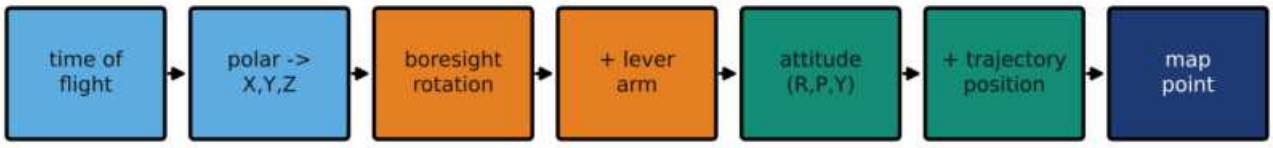


Figure 9.2: The chain of operations from a laser echo to a map coordinate.

SURVEYOR’S FIELD TIP

In ROS-based systems the lever arm and boresight live in the static transform `/tf_static` between `imu_link` and the LiDAR frame, e.g. translation $[0.120,\ 0.000,\ 0.085]$ m and roll-pitch-yaw $[0.11^\circ,\ -0.23^\circ,\ 0.01^\circ]$. If that transform is missing or wrong, you get the bird’s nest of Module 1.

9.3 How boresight is calibrated

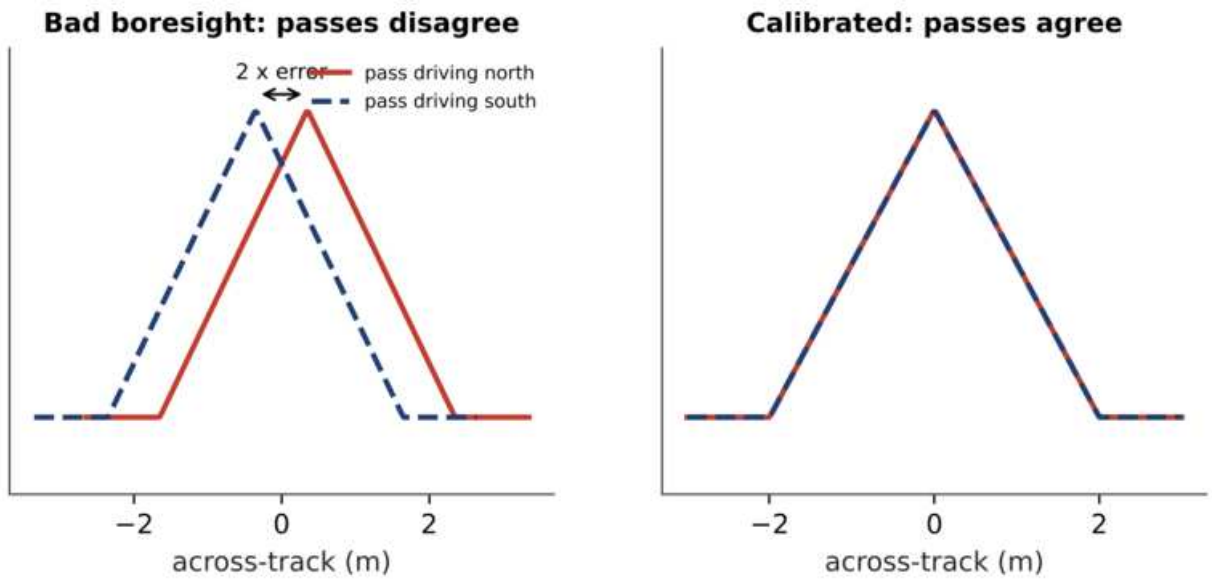


Figure 9.3: Driving past the same building in opposite directions doubles the visible effect of a boresight error.

SURVEYOR'S FIELD TIP

Treat calibration like any instrument adjustment: re-check after any mount change, impact, or long storage, and at the start of every project with a short loop over known features. Document the values used for every dataset.

MODULE 10

SLAM: Mapping While Finding Yourself

10.1 The chicken-and-egg problem

SIMPLE APPROACH

You are blindfolded in a new house and must draw a map of it. To draw a wall in the right place, you need to know where you are standing. But to know where you are standing, you need a map! SLAM is the clever trick of doing both a little at a time: take a step, feel a wall, guess where you are, add the wall to your map, take another step... and when you bump into a wall you already drew, you fix up your whole map.

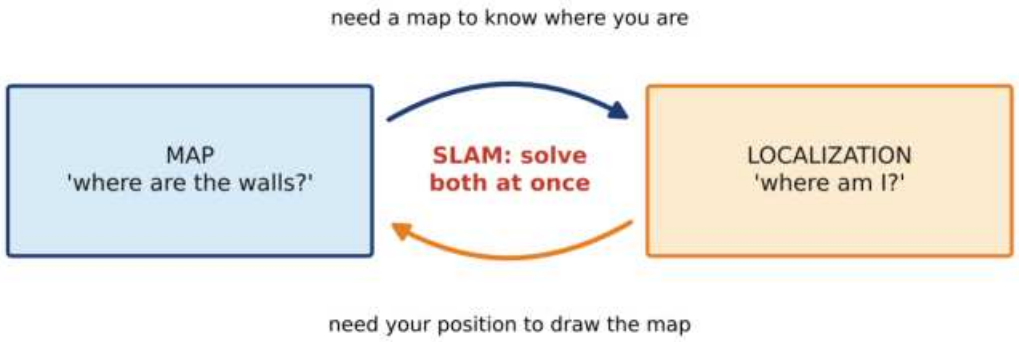


Figure 10.1: Simultaneous Localization And Mapping solves both questions together.

10.2 The motion model

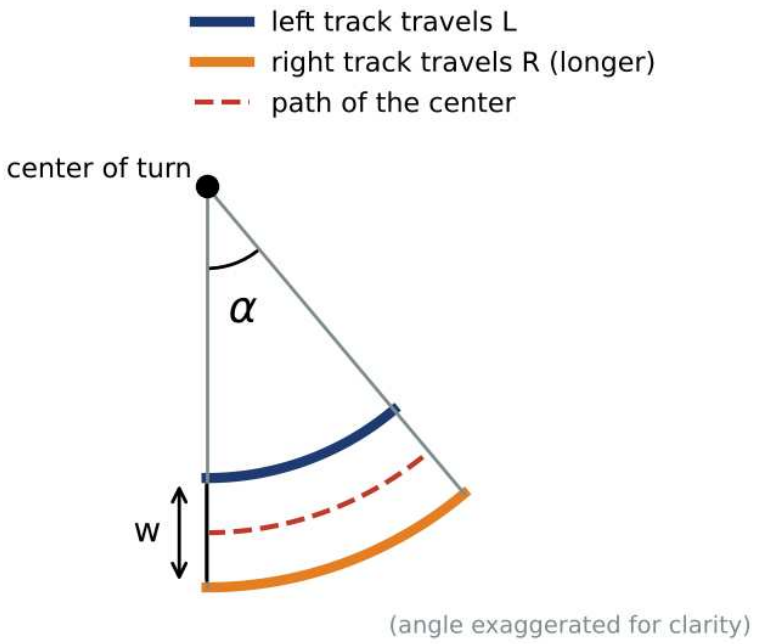


Figure 10.2: A two-track (differential drive) platform: when one side travels farther, the platform turns.

10.3 Uncertainty grows when you move and shrinks when you look

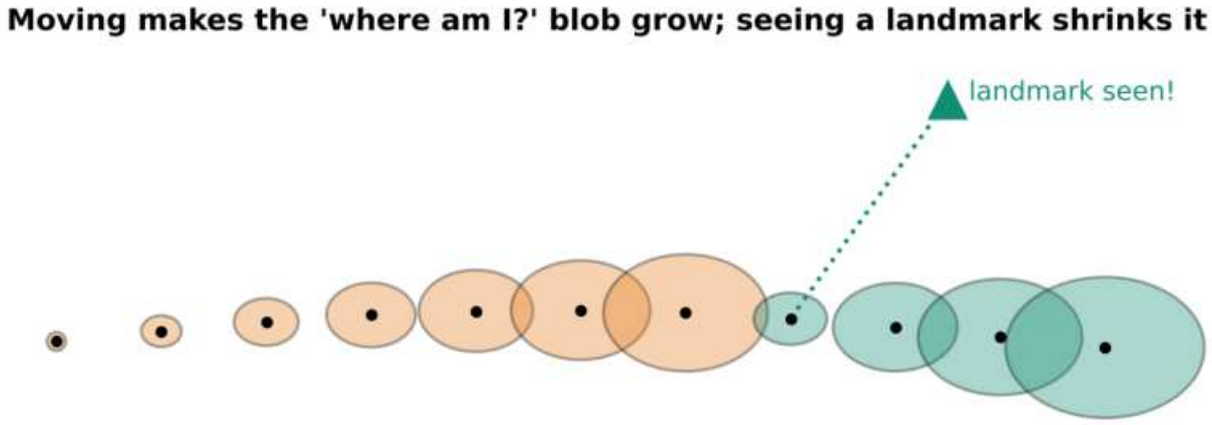


Figure 10.3: Each prediction inflates the 2σ uncertainty ellipse; one good landmark observation collapses it.

10.4 The Kalman filter in one dimension

SIMPLE APPROACH

Two friends guess how tall a tree is. One is a careful measurer; the other is just eyeballing. You trust the careful one more, so your final answer is *between* their guesses but closer to the careful one. The Kalman filter does exactly that, with numbers that say *how much* closer.

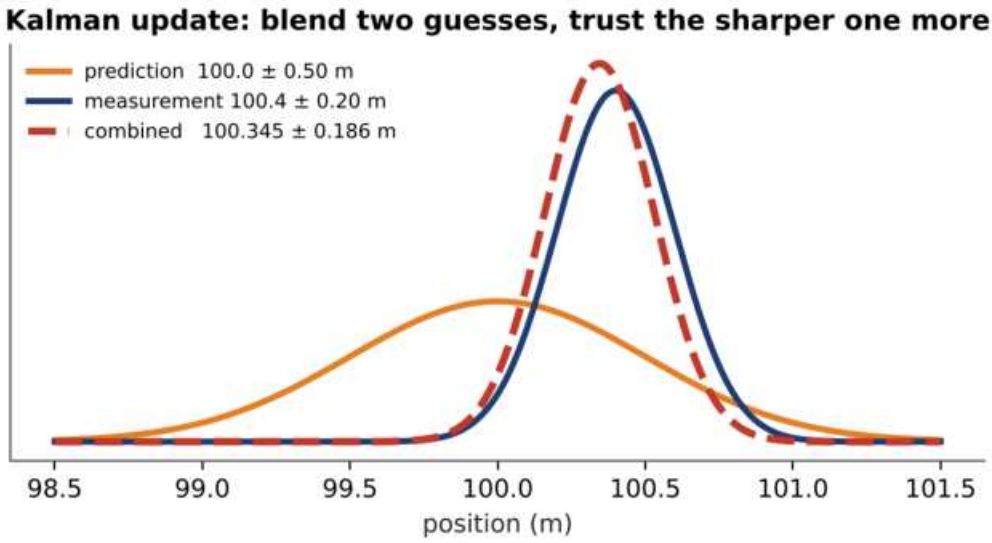


Figure 10.4: A wide prediction and a sharp measurement combine into an even sharper estimate.

10.5 The extended Kalman filter (EKF)

10.5.1 Correcting with a landmark

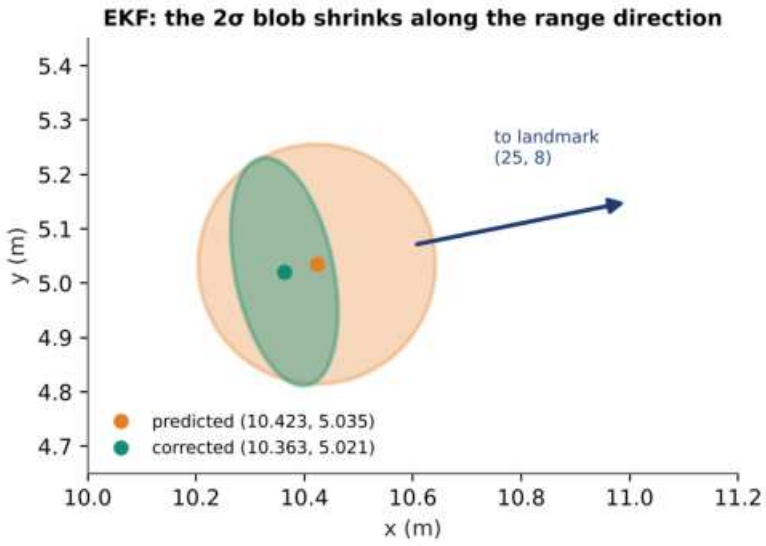


Figure 10.5: The 2σ ellipse before (orange) and after (teal) seeing the landmark.

10.6 Particle filters and EKF-SLAM

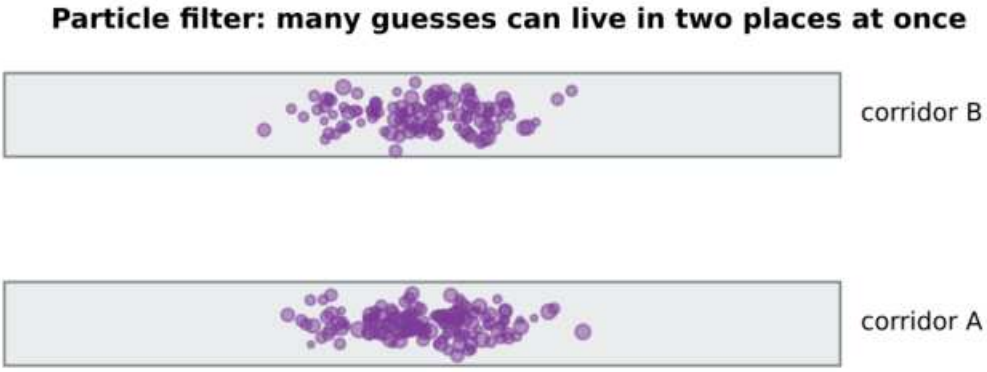


Figure 10.6: Particles can represent "I'm in corridor A *or* corridor B."

10.7 Drift and loop closure

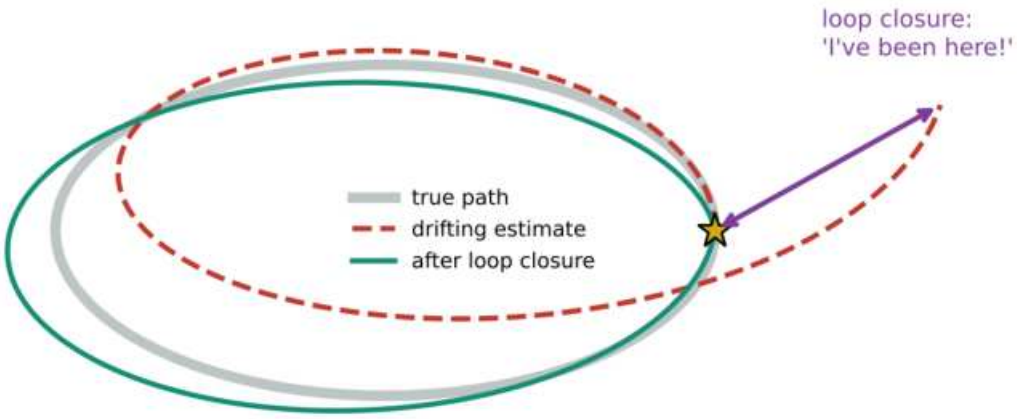


Figure 10.7: Odometry drifts like an open traverse; recognizing the start point closes the loop and distributes the misclosure.

SURVEYOR'S FIELD TIP

Plan routes like traverses: return to the start, cross earlier paths, and add cross-ties in long corridors. Every revisit is a free closure. A route that never revisits anything is an open traverse — its end can be far off and nothing in the data will tell you.

MODULE 11

Scan Matching: Fitting Clouds Together

11.1 The puzzle-piece idea

SIMPLE APPROACH

Draw a shape on a see-through sheet. Your friend draws the same shape on another sheet, but a little shifted and turned. Lay the sheets on top of each other and slide and twist the top one until the drawings line up perfectly. How far you slid and twisted it tells you how your friend's sheet was placed! A LiDAR does this with two scans: sliding and twisting the new scan until it lines up with the old one tells it how far the car moved.

11.2 Iterative Closest Point (ICP)

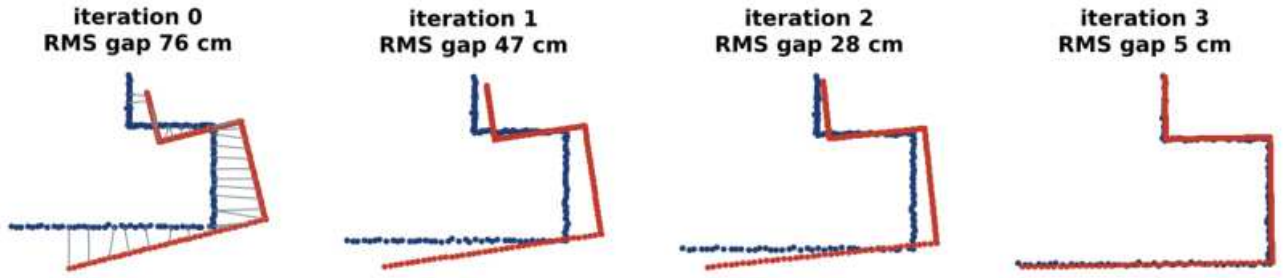


Figure 11.1: A real ICP run: pair each red point with its nearest blue point, solve the best fit, move, repeat.

11.3 Better cost functions: point-to-plane and NDT

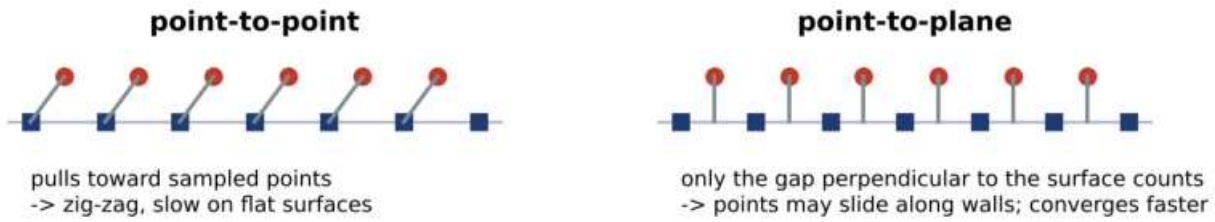


Figure 11.2: Point-to-point pulls toward sample points; point-to-plane only penalizes the gap normal to the surface.

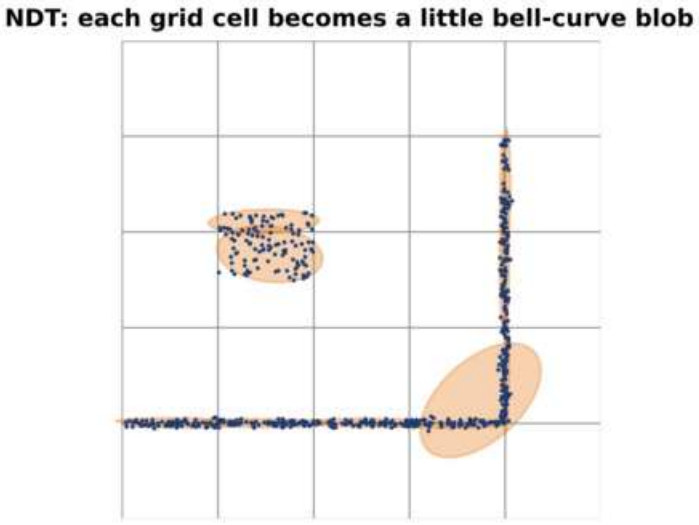


Figure 11.3: The Normal Distributions Transform summarizes each grid cell by a mean and covariance.

11.4 Feature-based matching (LOAM and LIO-SAM)

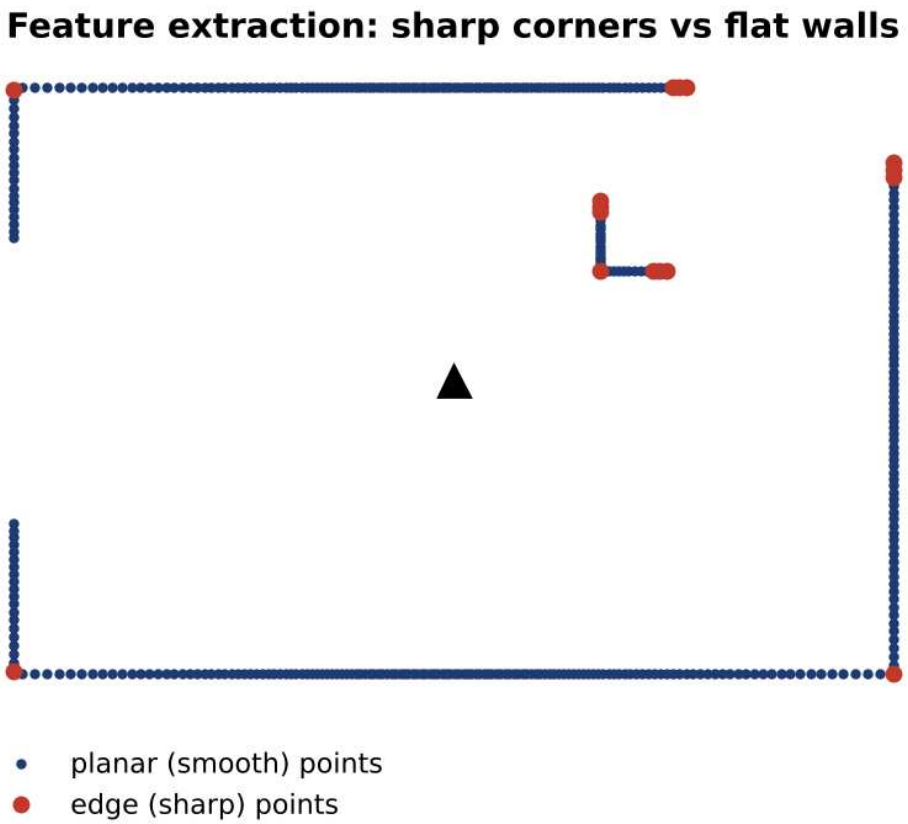


Figure 11.4: Points on sharp corners become edge features; points on smooth surfaces become planar features.

SIMPLE APPROACH

Instead of matching every single dot, look for the *special* ones: sharp corners (like the corner of a box) and flat sides (like a table top). Corners are easy to recognize and flat sides are easy to slide against. Matching only these is much faster.

11.4.1 Distances to edges and planes (LIO-SAM eqs. 10–11)

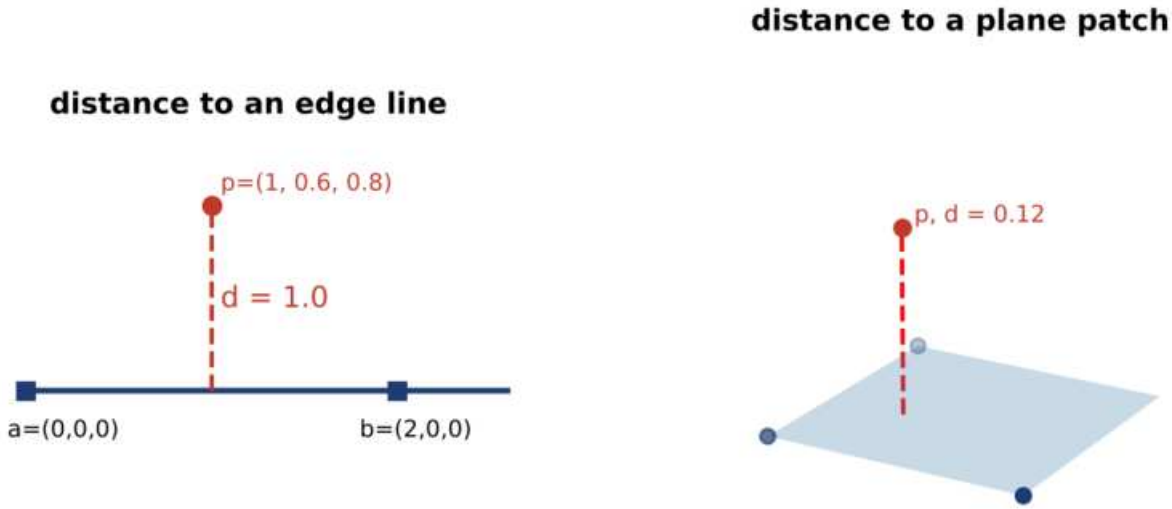


Figure 11.5: An edge feature is compared with a line through two map points; a planar feature with a plane through three.

11.5 When matching fails: the hallway effect

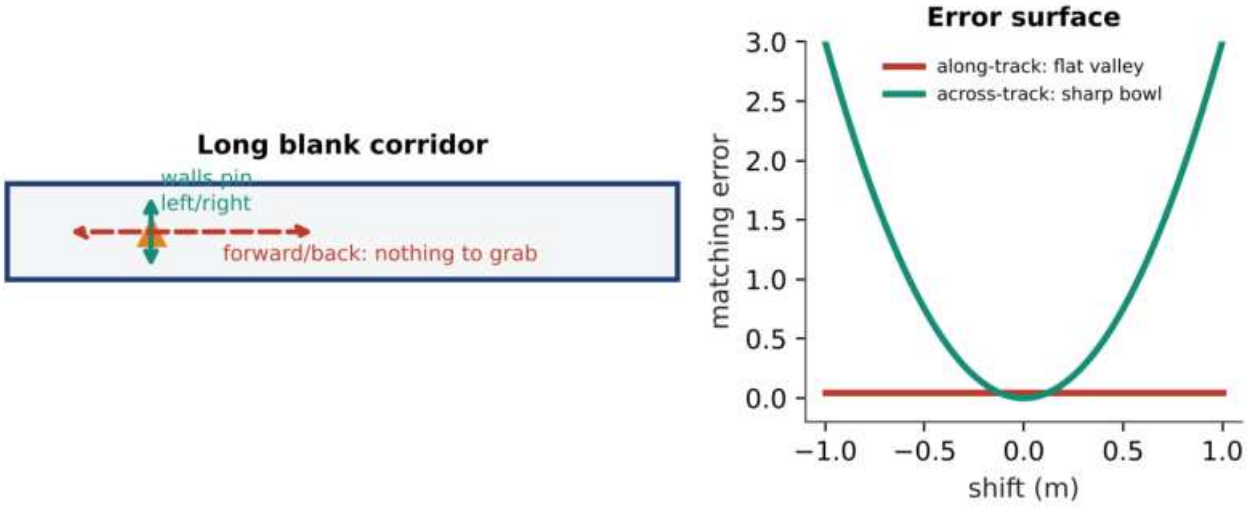


Figure 11.6: In a featureless corridor the matching error has a flat valley along the corridor — any forward shift fits equally well.

rooms or cross-ties to create unique geometry, place temporary objects (cones, boxes) if needed, and always check along-track distances against a taped or total-station measurement.

MODULE 12

Factor Graphs and LIO-SAM

12.1 From a filter to a graph

SIMPLE APPROACH

Imagine a row of people connected by rubber bands. Each rubber band says "you two should be this far apart." One person is also tied to a post that says "you belong here." If one band pulls a little wrong, everyone wiggles until the pulling is shared out as fairly as possible. A **factor graph** is that picture: the people are the car's positions over time, and the rubber bands are measurements.

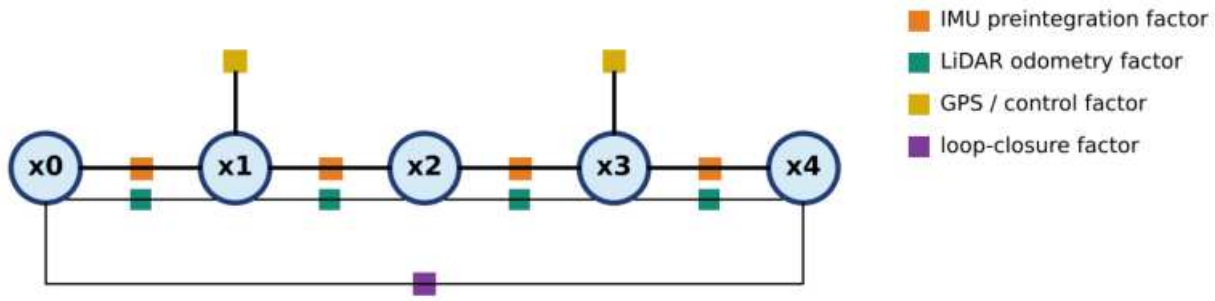


Figure 12.1: A factor graph: circles are unknown states (poses); small squares are measurements (factors) that tie them together.

12.2 A factor graph you can solve by hand

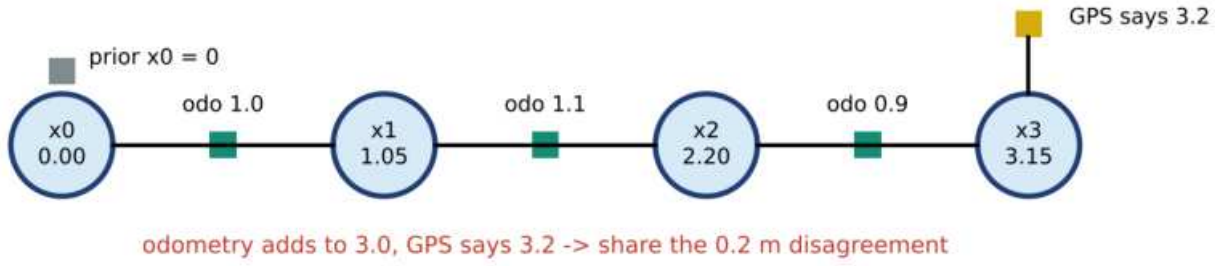


Figure 12.2: Four positions along a straight road, three odometry factors, one GPS factor and a prior.

12.3 LIO-SAM, piece by piece

12.3.1 Keyframes and the sliding window

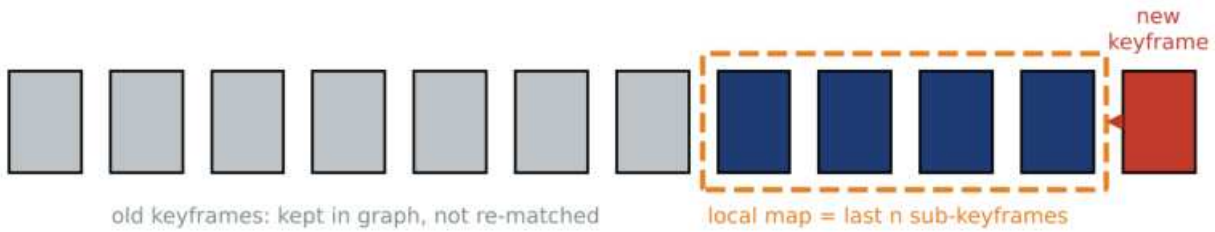


Figure 12.3: Only keyframes enter the graph; a new keyframe is matched to a local map of the last n sub-keyframes.

12.3.2 What it achieved

SURVEYOR'S FIELD TIP

You do not need to write an optimizer, but you should be able to answer: Which factors were active in this dataset? Where were the loop closures? Where was GNSS used? Where did the corridor degrade LiDAR odometry? Good processing software lets you view the trajectory colored by its estimated standard deviation — inspect it before trusting any part of the cloud.

12.4 Building the trajectory: poses, scans, IMU, GNSS and images

SIMPLE APPROACH

Think of a trip recorded as a flip-book. Each page is a **pose**: where the car was and which way it was pointing at one instant. The IMU writes a new page hundreds of times a second, so the flip-book is smooth — but it slowly drifts. The LiDAR compares each new scan with the last ones and corrects the drift, like recognizing the same buildings. GNSS now and then says "this page is exactly here on Earth." The camera takes pictures along the way, and each picture is glued to the page for the moment it was taken. Flip through all the pages and you have the **trajectory**: the path, with direction, at every instant.

The words

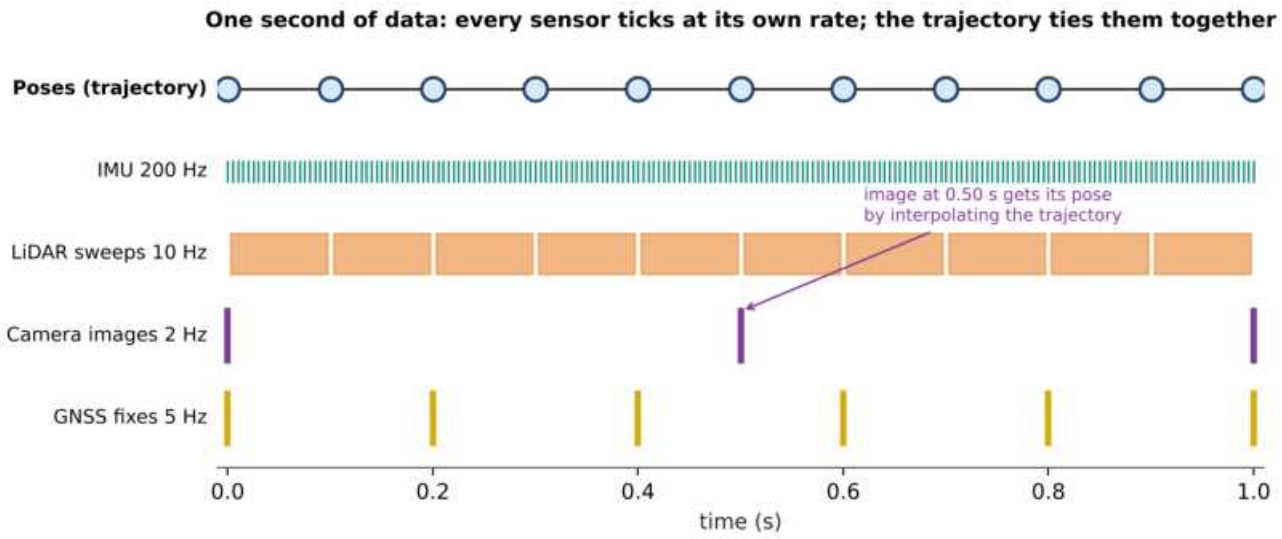


Figure 12.4: One second of data. The IMU ticks 200 times, the LiDAR completes 10 sweeps, GNSS gives 5 fixes and the camera takes 2 images. The trajectory (top row) is the common thread that gives every one of them a pose.

What each sensor contributes

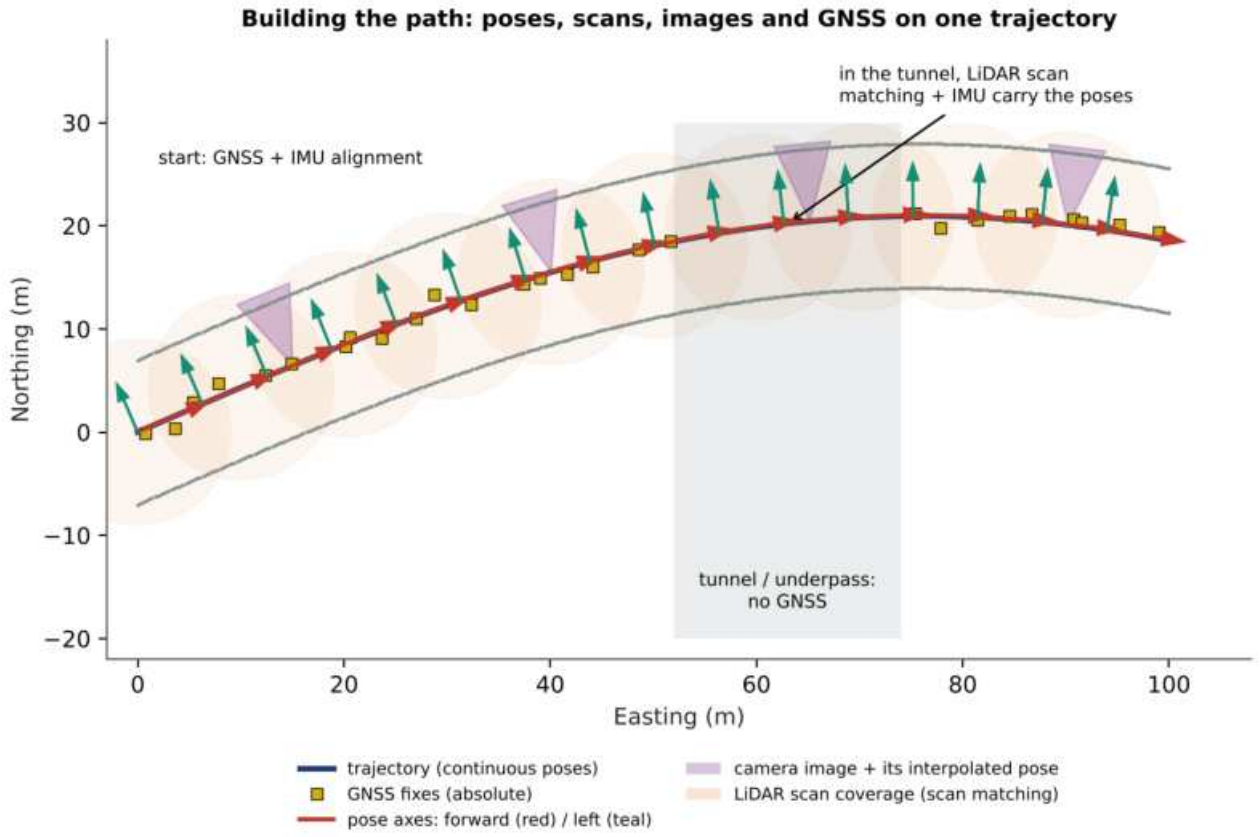


Figure 12.5: The path as built: blue trajectory with pose axes, orange LiDAR scan coverage used for matching, gold GNSS fixes (absent in the tunnel), and purple camera images, each placed with its own interpolated pose.

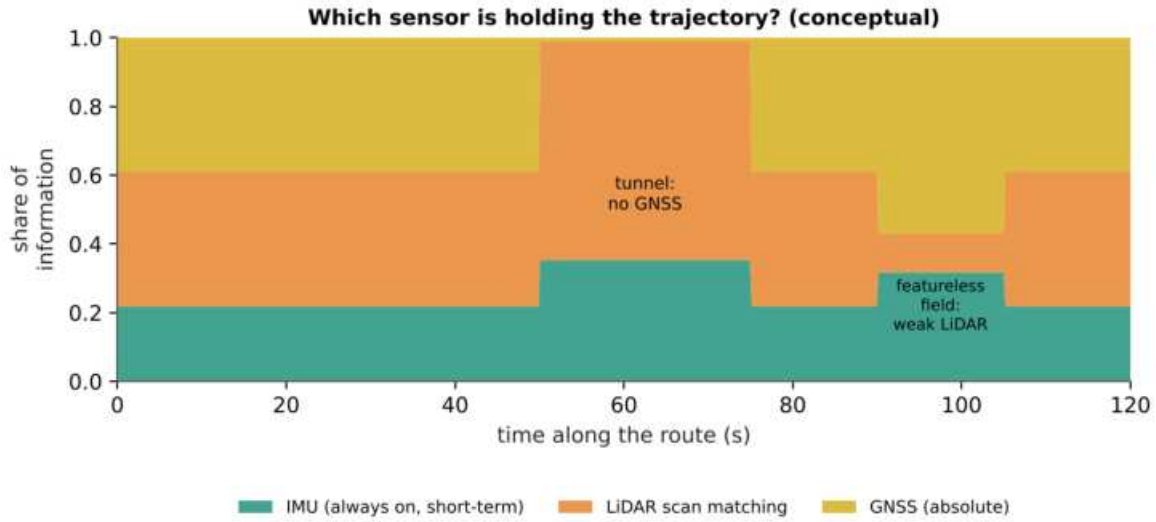


Figure 12.6: Conceptually, the factor graph leans on whichever sensors have information at each moment: LiDAR and IMU carry the tunnel; GNSS and IMU carry the featureless field.

SURVEYOR'S FIELD TIP

Deliver and archive the trajectory. Keep the final trajectory (time, position, attitude and their standard deviations), the image exterior orientations, the calibration file and the processing log with the project. When a client later asks "where did this point come from?", the trajectory is the answer — and it is the first thing a reviewer will inspect for drift, gaps and GNSS outages.

MODULE 13

Control, Error Budgets and Accuracy Standards

13.1 Relative versus absolute

SIMPLE APPROACH

You can draw a perfect map of your bedroom — every toy in the right place compared to every other toy — but if you don't know where your house is on the street, nobody else can use it to find your room. SLAM makes the bedroom map. **Control points** — spots measured very carefully by surveyors — pin that map onto the real world. **Checkpoints** are extra careful spots that we keep secret from the computer, then use to test whether the map is truly right.

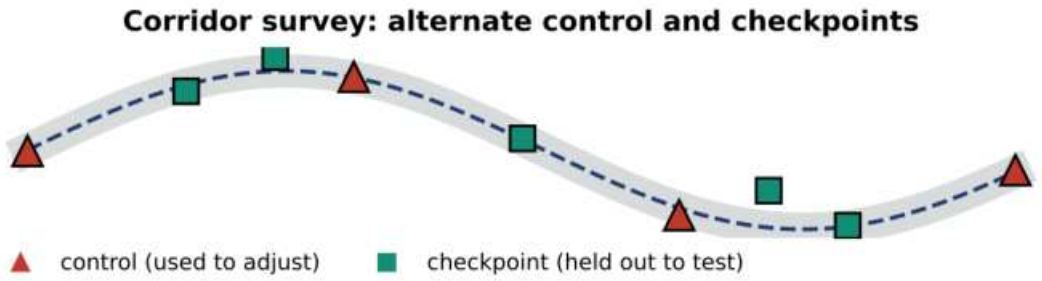


Figure 13.1: Along a corridor, alternate points used to constrain the solution (control) with points held back purely to test it (checkpoints).

SURVEYOR'S FIELD TIP

- Use targets the scanner can see and the software can measure: painted chevrons or checkerboards on pavement for vertical and horizontal checks; high-intensity paint shows clearly in intensity.
- Survey control and checkpoints with a method at least **two times** more accurate than the product you are testing (ASPRS Edition 2), and include their own uncertainty in the final accuracy (below).
- Distribute checkpoints along the whole project and across the cross-section — not only at the ends near GNSS-friendly areas.
- Confirm the vertical datum and geoid model match project requirements before comparing anything.

13.2 Computing accuracy from checkpoints

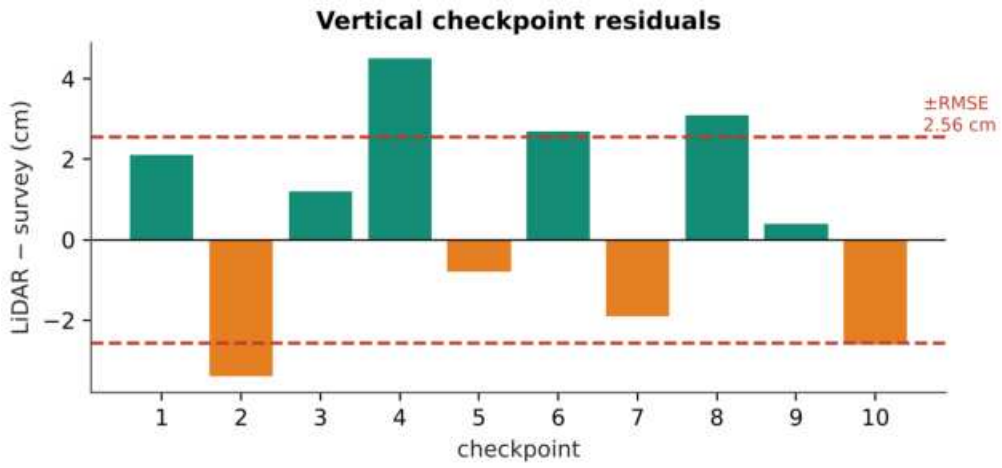


Figure 13.2: Vertical residuals (LiDAR surface minus surveyed elevation) at ten checkpoints.

13.3 ASPRS lidar data standards in detail

SIMPLE APPROACH

One rulebook says *how close to the truth* the dots must be and how to check them with secret checkpoints. The other rulebook says *how to write the dots down* in a file, including a label on every dot: ground, building, tree, water, wire, noise… so any software can read it.

MODULE 14

Error Propagation and Error Budgets

14.1 Three kinds of error

SIMPLE APPROACH

Imagine throwing darts. If your darts land all over the board, that is **random error** — your hand shakes a little differently every time. If all your darts land in a tight group, but always 5 cm to the left of the bullseye, that is **systematic error** — maybe the board is crooked. If one dart hits the wall, that is a **blunder**. You fix each one in a different way: practice to shrink random error, straighten the board to remove systematic error, and throw out the dart that hit the wall.

14.2 The law of propagation of variances (one output)

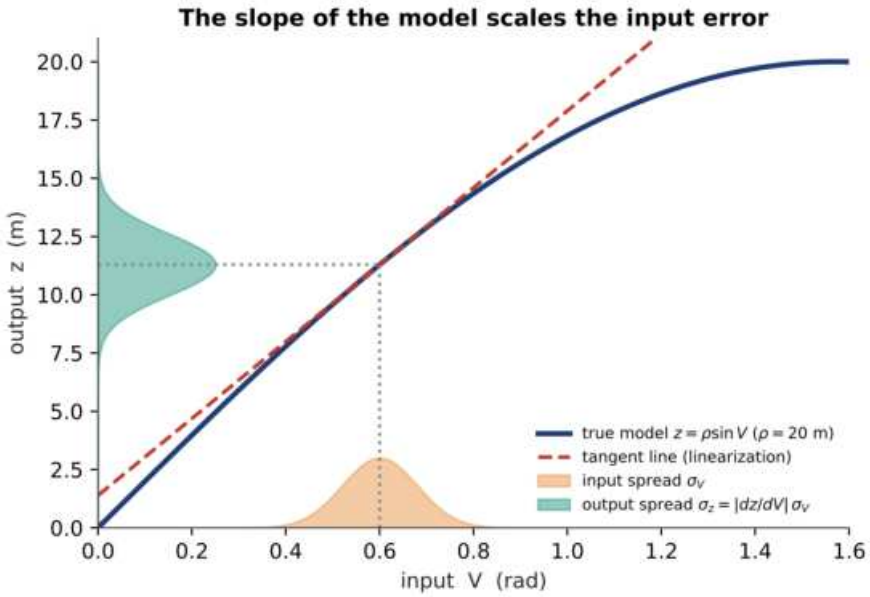


Figure 14.1: Near the operating point the model is almost a straight line. Its slope converts input spread into output spread.

14.3 When errors are correlated

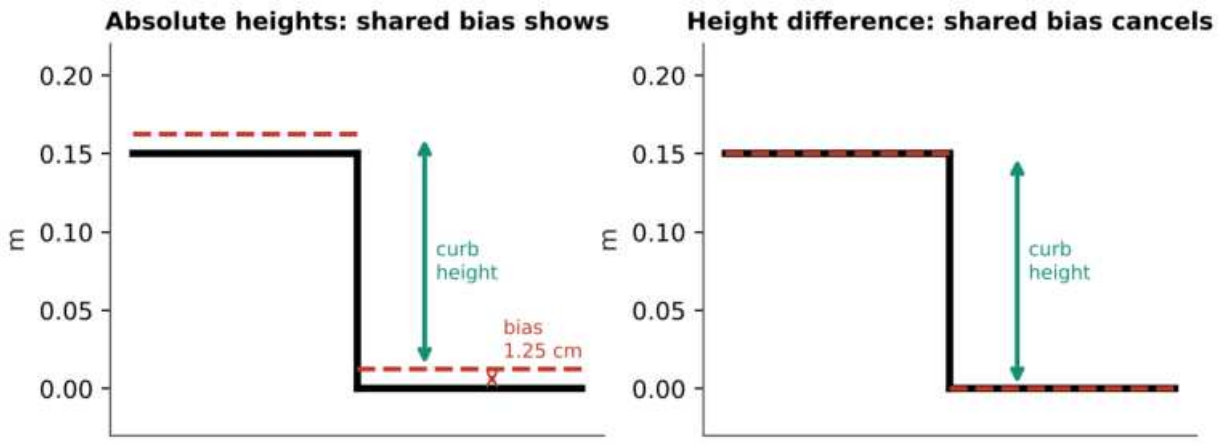


Figure 14.2: A bias shared by both points shifts absolute heights but disappears from a height *difference*.

14.4 Many outputs at once: the covariance matrix

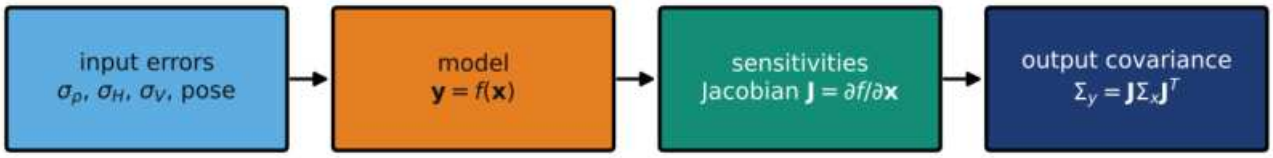


Figure 14.3: The general propagation recipe.

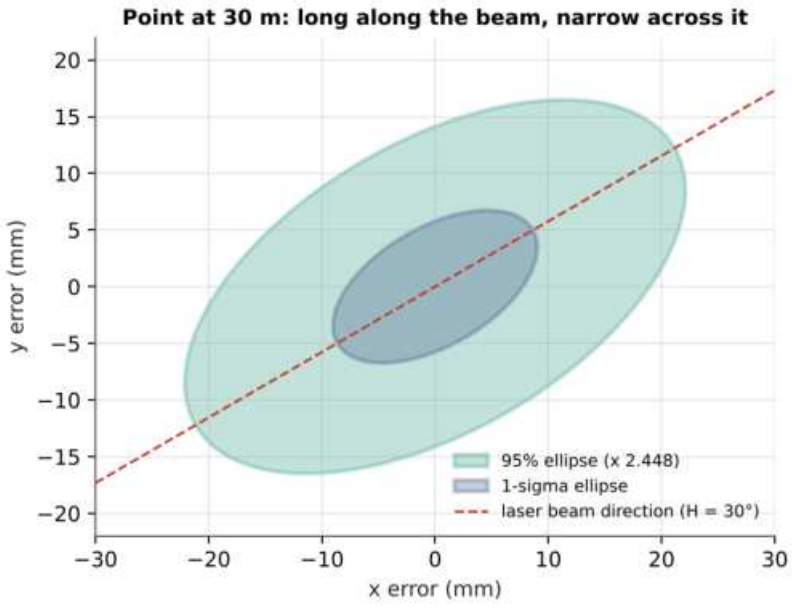


Figure 14.4: Horizontal error ellipse of the worked example. The 95% ellipse is 2.448 times the 1σ ellipse.

14.5 Propagating through the full georeferencing equation

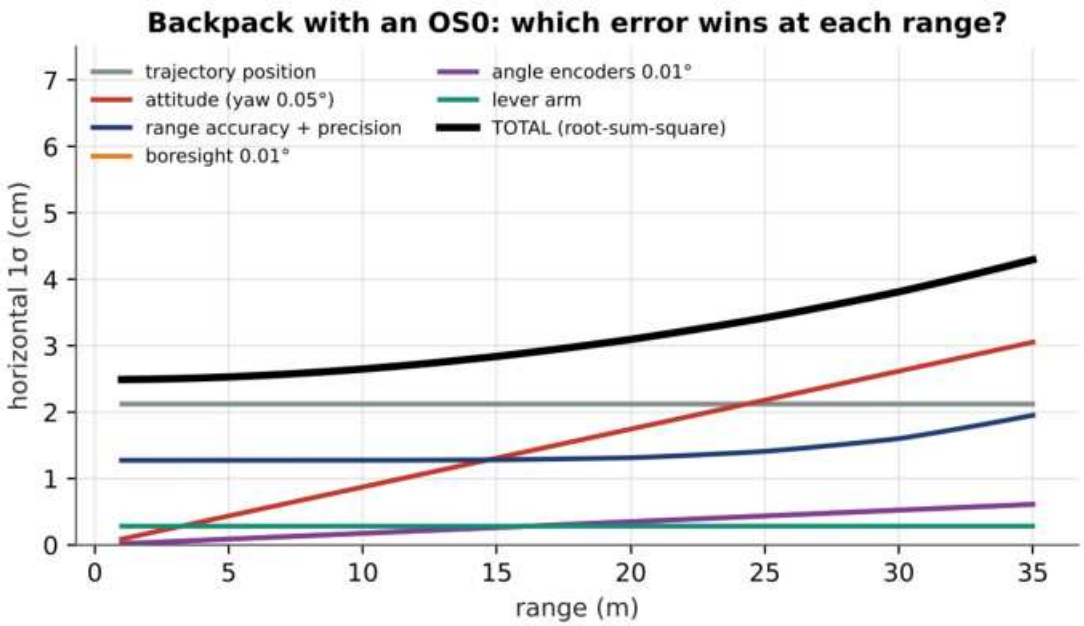


Figure 14.5: Contributions versus range for the same backpack. Fixed terms dominate close in; angle terms take over far out.

SURVEYOR'S FIELD TIP

Use the curve to set your **maximum useful range**. With this system, a 3 cm horizontal spec is met only out to about 18 m; beyond that, either filter points by range in processing or improve the heading (more loop closures, better IMU, control along the route).

14.6 Checking with Monte Carlo simulation

SIMPLE APPROACH

Instead of doing hard math, you can just *pretend* thousands of times. Roll dice to make up small random errors for every input, compute the point, and write it down. Do it 100,000 times on a computer and look at how spread out the answers are. If the spread matches the formula, you know the formula was right.

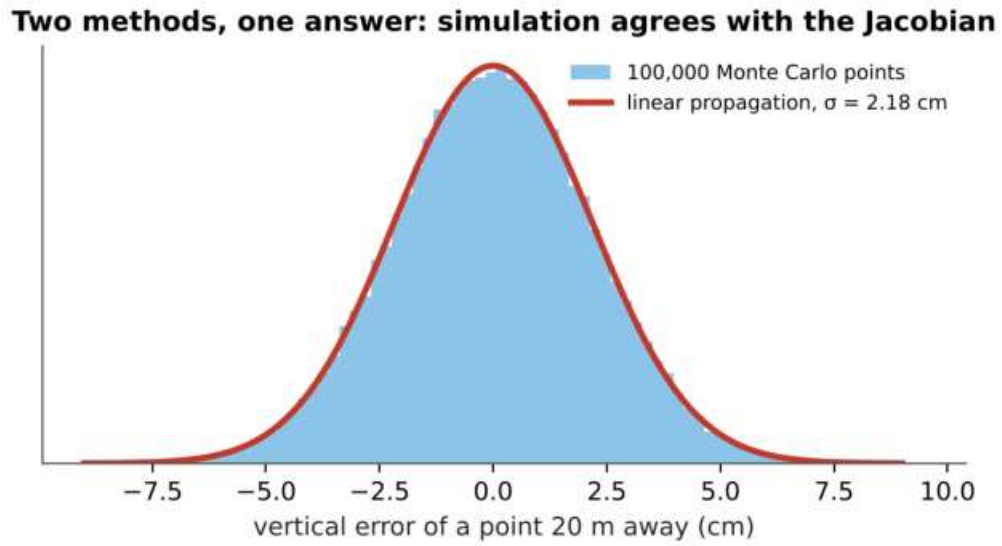


Figure 14.6: Vertical error of the 20 m point: 100 000 simulated georeferencings (bars) versus the linear-propagation Gaussian (line).

14.7 Averaging, and the floor set by bias

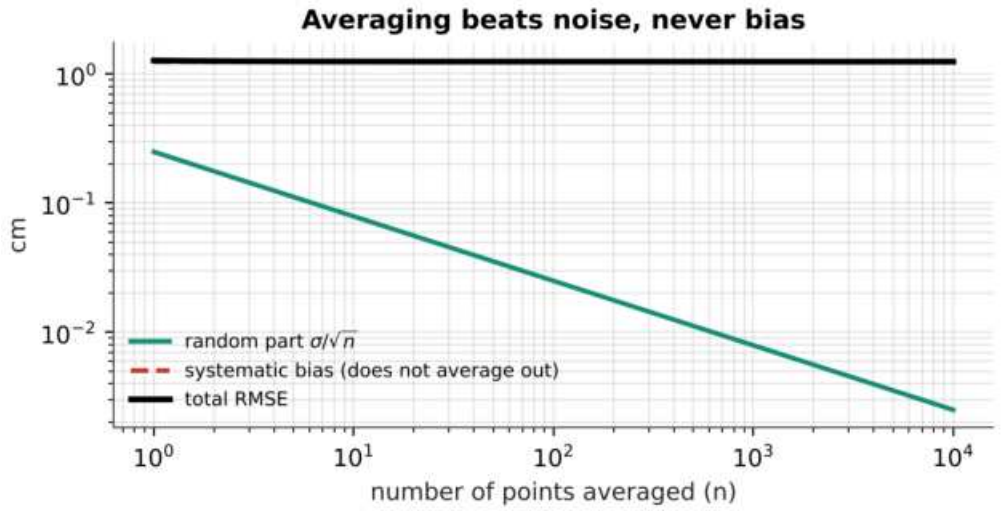


Figure 14.7: Averaging n points shrinks random error as $1/\sqrt{n}$ but cannot go below the systematic bias.

14.8 Error budgets: from sources to a total

SIMPLE APPROACH

If you pack a backpack, every item takes up some room, and the bag can only hold so much. An error budget is a list of every little thing that can make a dot land in the wrong place, and how much "room" each one takes, so we know the whole bag still closes.

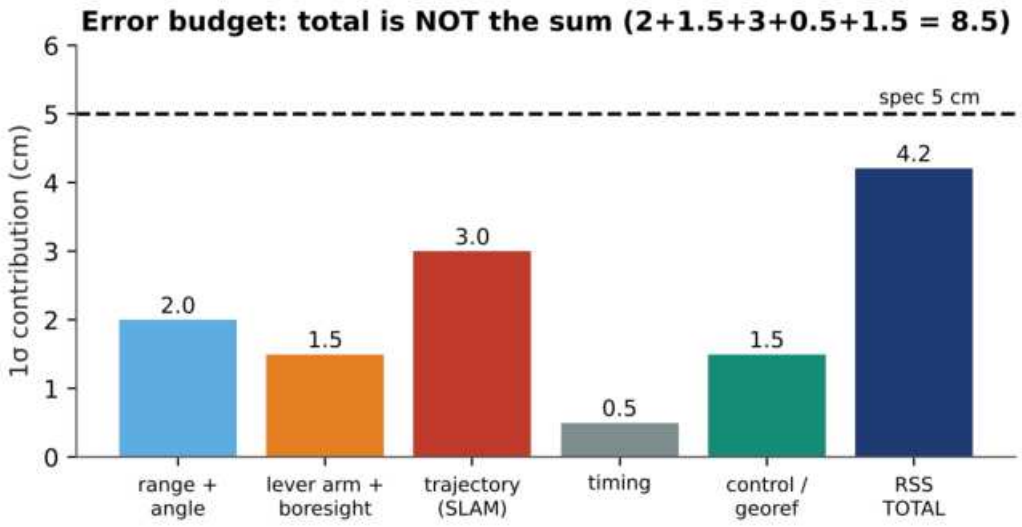


Figure 14.8: Independent 1σ contributions combine by root-sum-square, not by simple addition.

14.9 Error budgets for the case-study systems

14.9.1 IMU attitude: GX5-25, GX5-45 and GQ7 (Module 16)

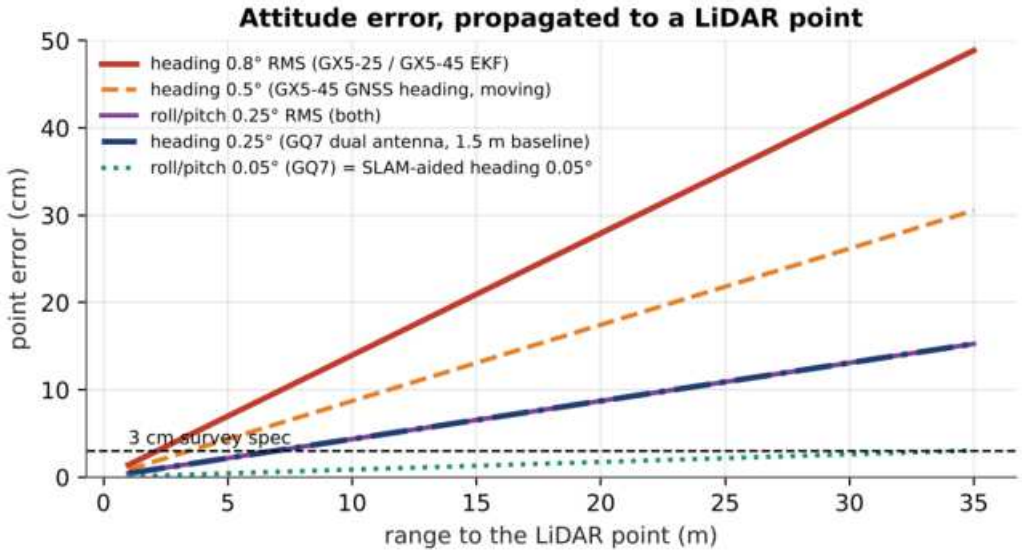


Figure 14.9: Sideways and vertical point error caused by each sensor's attitude accuracy, versus range. The GQ7 heading line coincides with the GX5 roll/pitch line, and the GQ7 roll/pitch line with SLAM-aided heading.

14.9.2 RIEGL VUX-240[24] at 500 ft (Module 17)

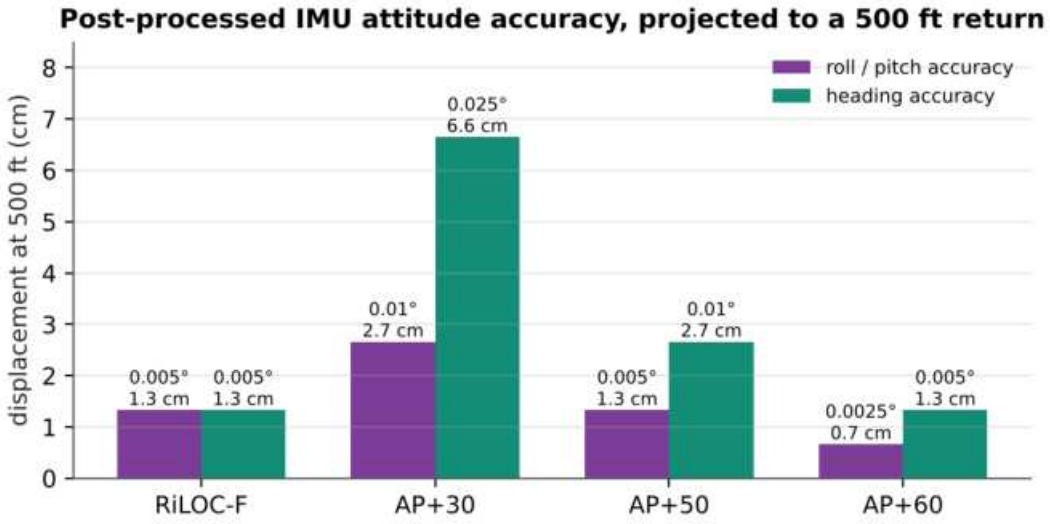


Figure 14.10: Each IMU option's post-processed attitude accuracy projected to the ground at 500 ft (small-angle rule, Module 2).

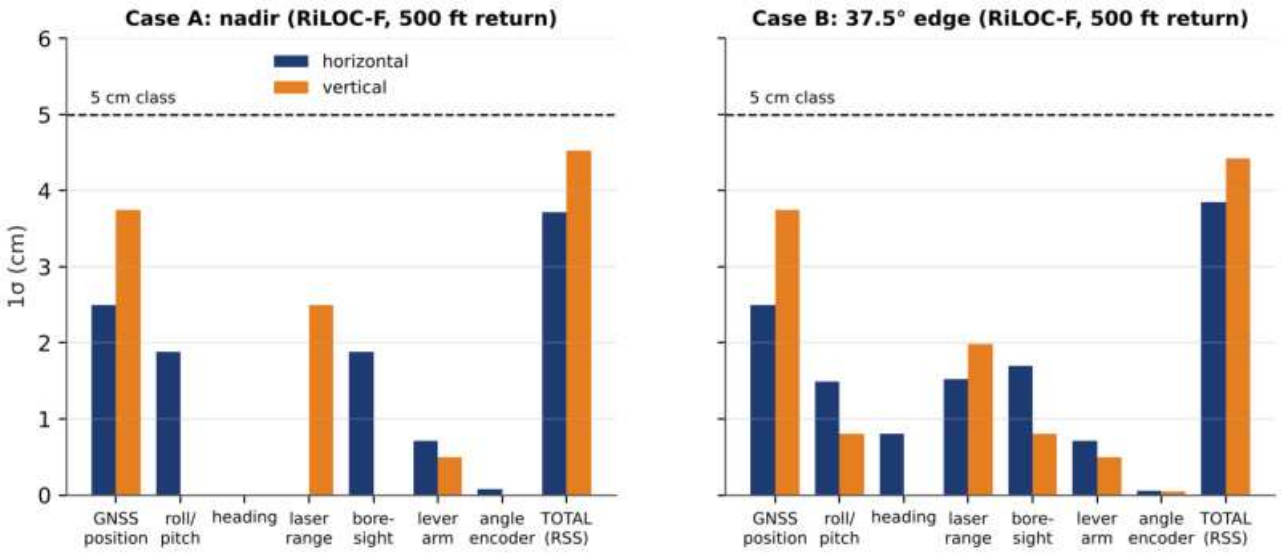


Figure 14.11: Error contributions (1σ) for the RiLOC-F at 500 ft. The dashed line is the 5 cm class.

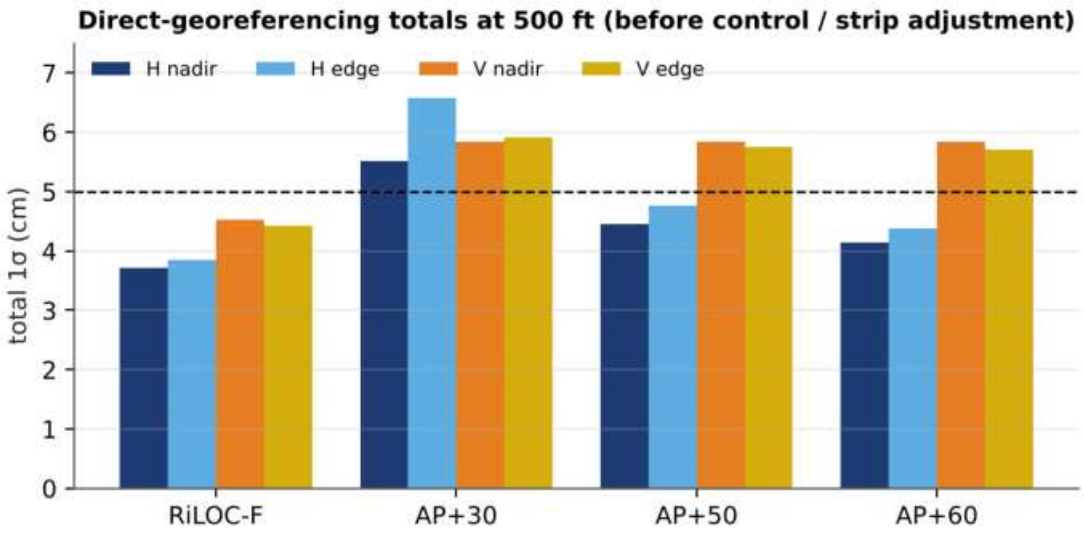


Figure 14.12: Totals for each IMU/GNSS option. Better attitude helps horizontal; vertical is limited by GNSS and laser range.

MODULE 15

Ouster OS0 Case Study: Errors and Resolution

SIMPLE APPROACH

A datasheet is like the label on a box of crayons: it tells you how many colors, how sharp the tips are, and how long they last. But to know whether you can draw a tiny ant or only a big elephant, you have to do a little math with those numbers. That is what this module does for the OS0 laser scanner.

15.1 Angular resolution and point spacing

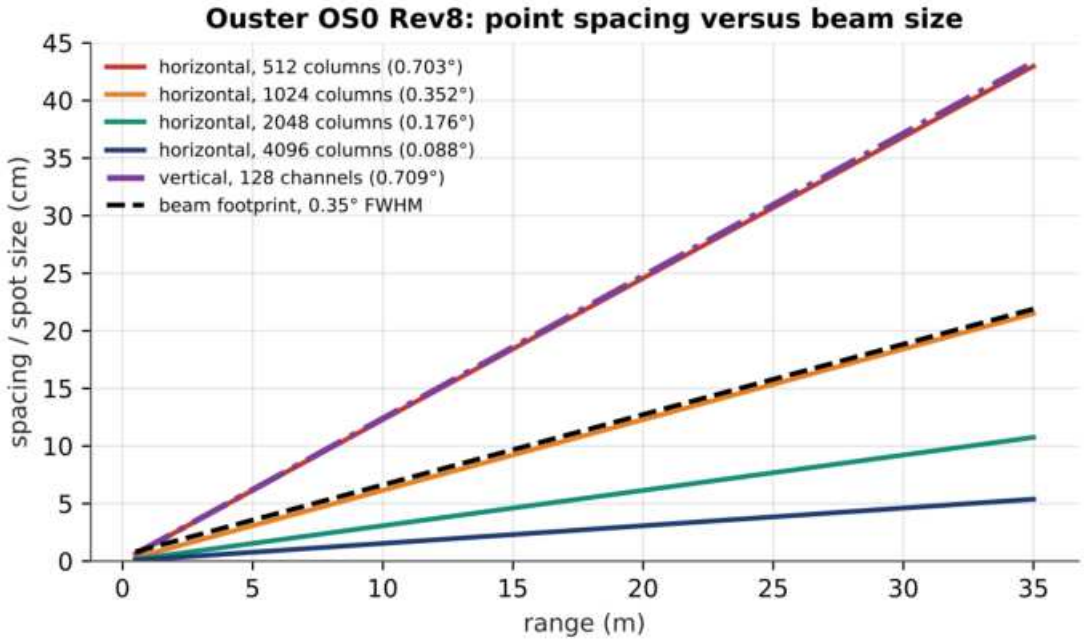


Figure 15.1: Point spacing for each OS0 mode, compared with the size of the laser spot.

15.2 Beam footprint: the real resolution limit

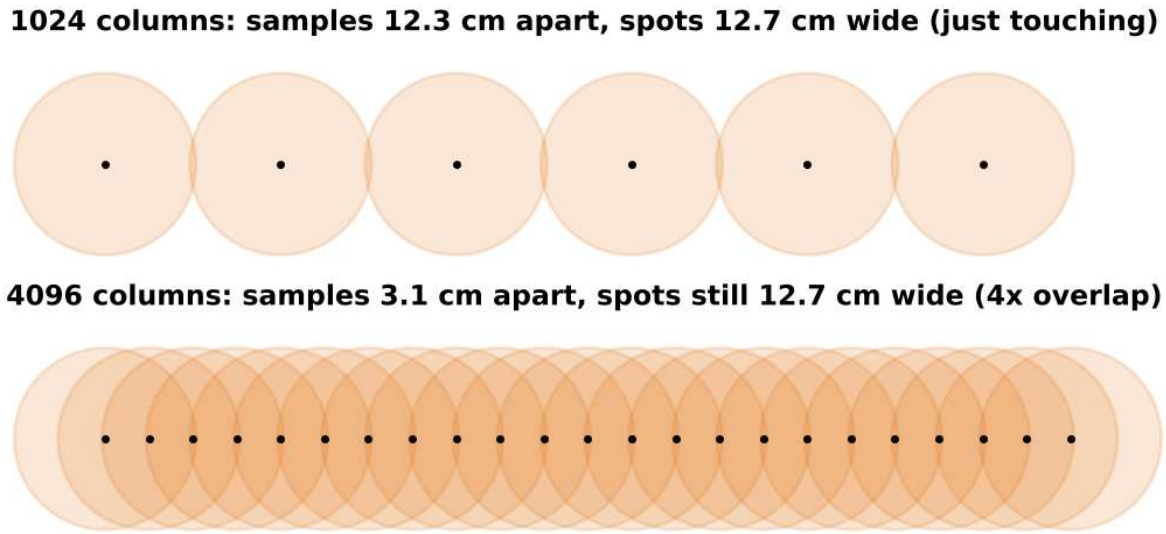


Figure 15.2: At 20 m, the 1024 mode places spots edge to edge; the 4096 mode overlaps each spot with its neighbors.

15.3 Range resolution versus precision versus accuracy

SIMPLE APPROACH

Think of a ruler. **Resolution** is how small the marks are (millimeters). **Precision** is how much your answer wiggles if you measure the same thing ten times. **Accuracy** is whether the ruler itself is the right length. A ruler can have tiny marks and still be stretched!

15.3.1 Precision and accuracy versus range and reflectivity

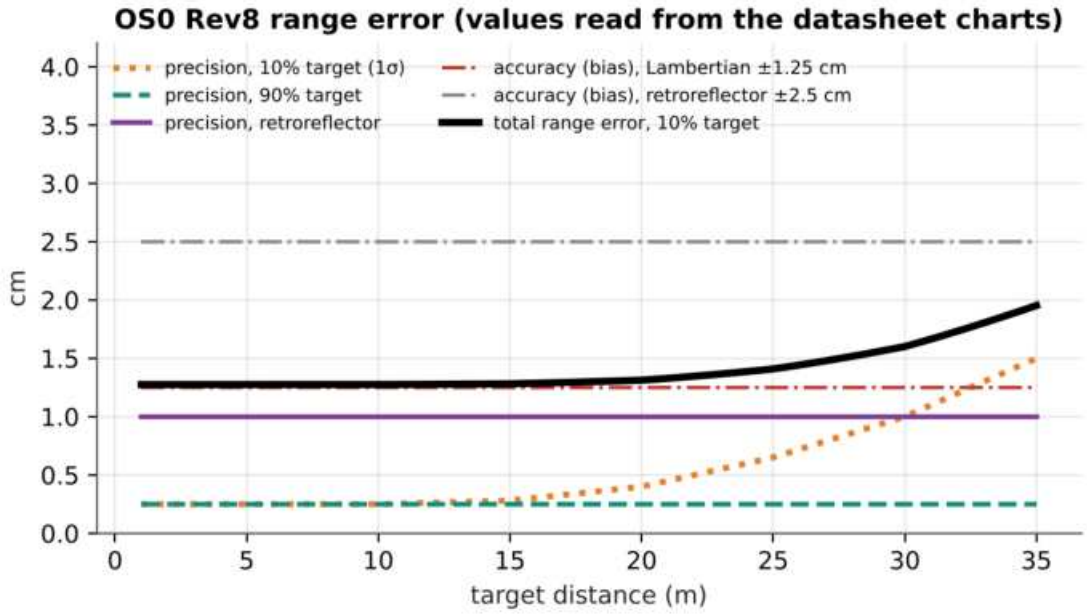


Figure 15.3: Range precision (1σ) and accuracy for the OS0 Rev8. Curves for the 10% target are read from the datasheet chart and are approximate.

SURVEYOR'S FIELD TIP

Retroreflective targets are a trap. The datasheet gives a 2.5 cm accuracy and 1.0 cm precision on retroreflectors — worse than on ordinary surfaces — because the very strong return distorts the pulse timing. Do not use retro-tape targets as checkpoints for absolute height unless the bias has been calibrated; use matte painted targets (Lambertian) and use retro targets for horizontal identification only.

15.4 Timing, synchronization and skew

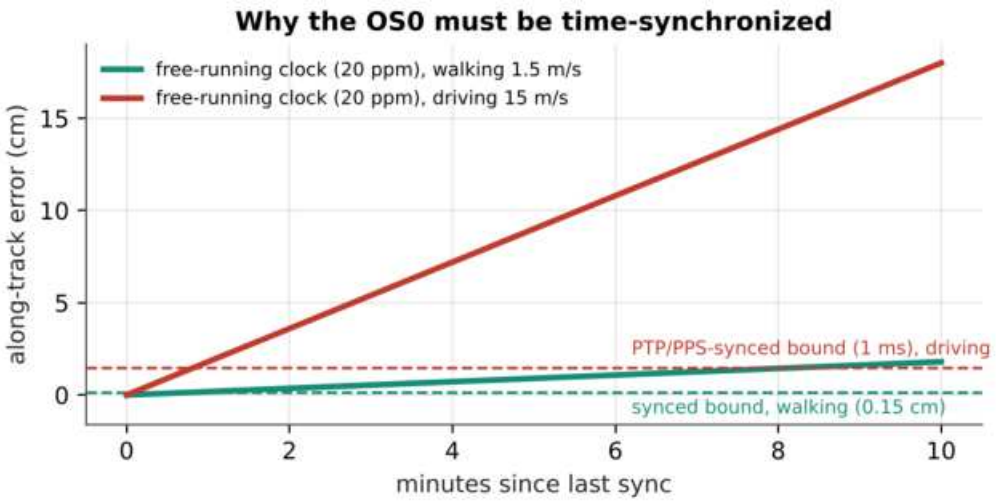


Figure 15.4: Along-track error from timing for a free-running OS0 clock versus a PTP/PPS-synchronized one.

MODULE 16

Choosing an IMU: AHRS, GNSS/INS and Tactical GNSS/INS

SIMPLE APPROACH

Imagine two friends walking around the block with their eyes closed. The first friend has a compass and counts steps. The second friend has the same compass and step-counter but also gets a phone message every second saying "you are about here, give or take two meters." Who gets home closest? And who can tell you exactly which way the mailbox is? That is the whole story of this module.

16.1 The Block Loop Challenge

Run 1 — AHRS heading + odometer, systematic errors only

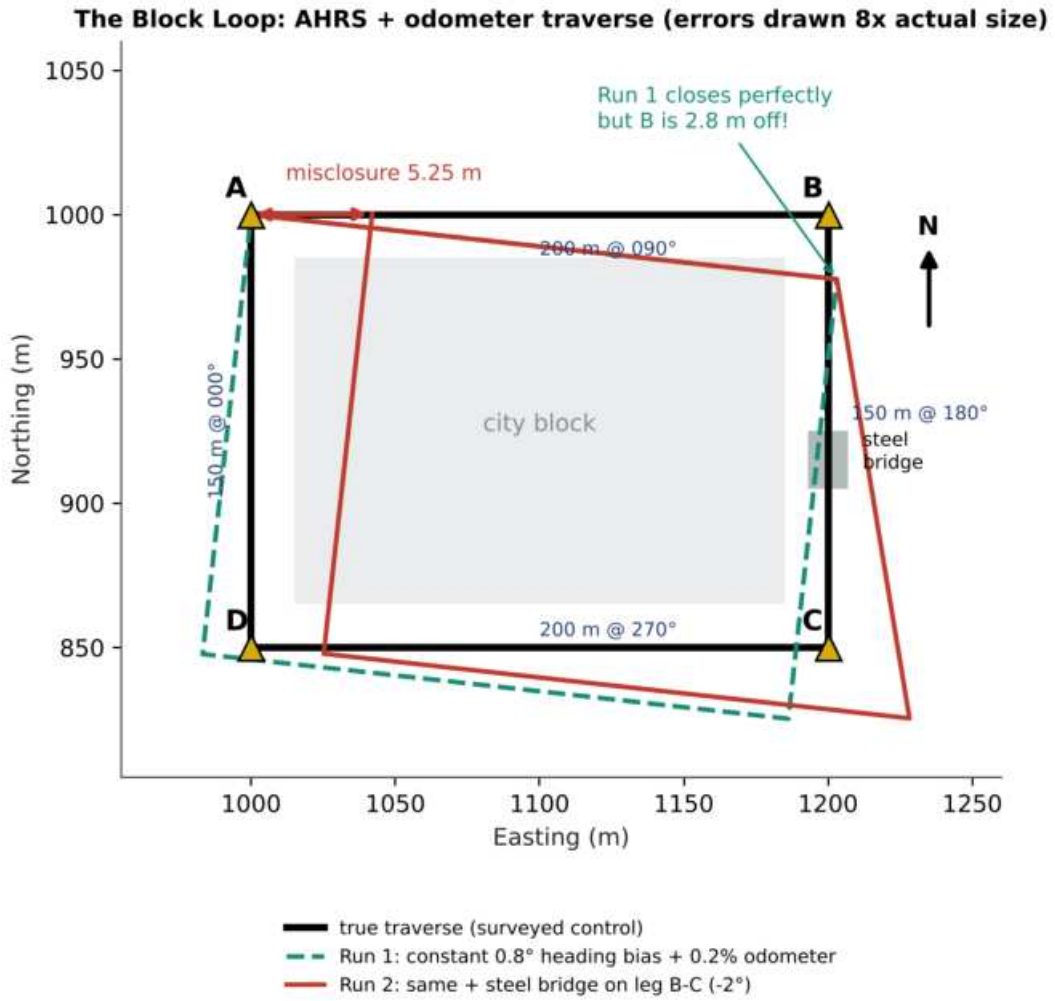


Figure 16.1: Sketch of the traverse with the AHRS runs. Errors are drawn 8 times larger than computed so they can be seen.

BIG IDEA

Closure tests catch errors that *change* around the loop. They are blind to errors that are the *same* on every leg.

Run 4 — pure inertial navigation with the GX5-25

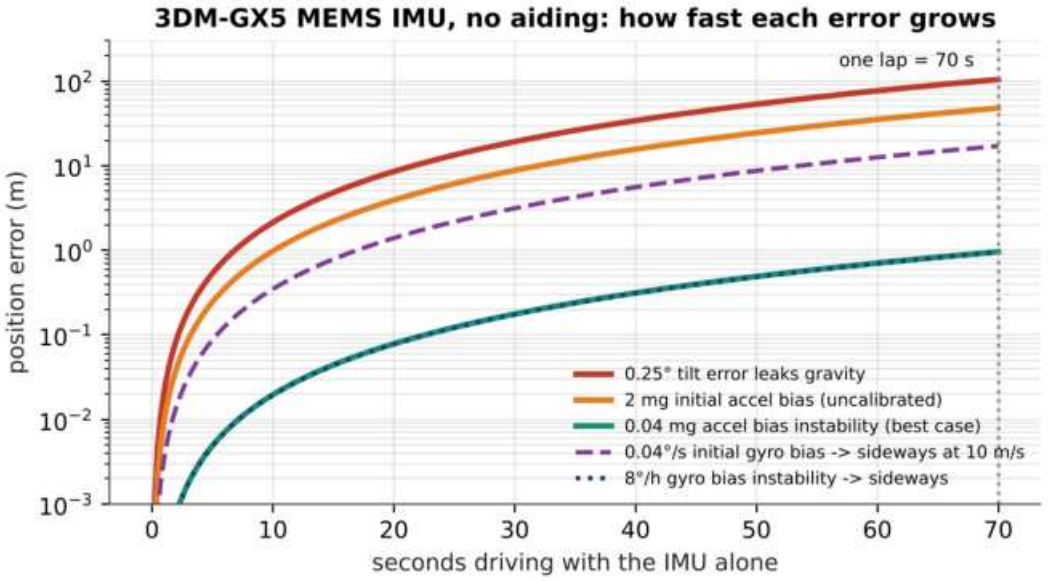


Figure 16.2: Each IMU error source, integrated with no aiding at all, over one 70 s lap.

Run 5 — the 3DM-GX5-45 GNSS/INS

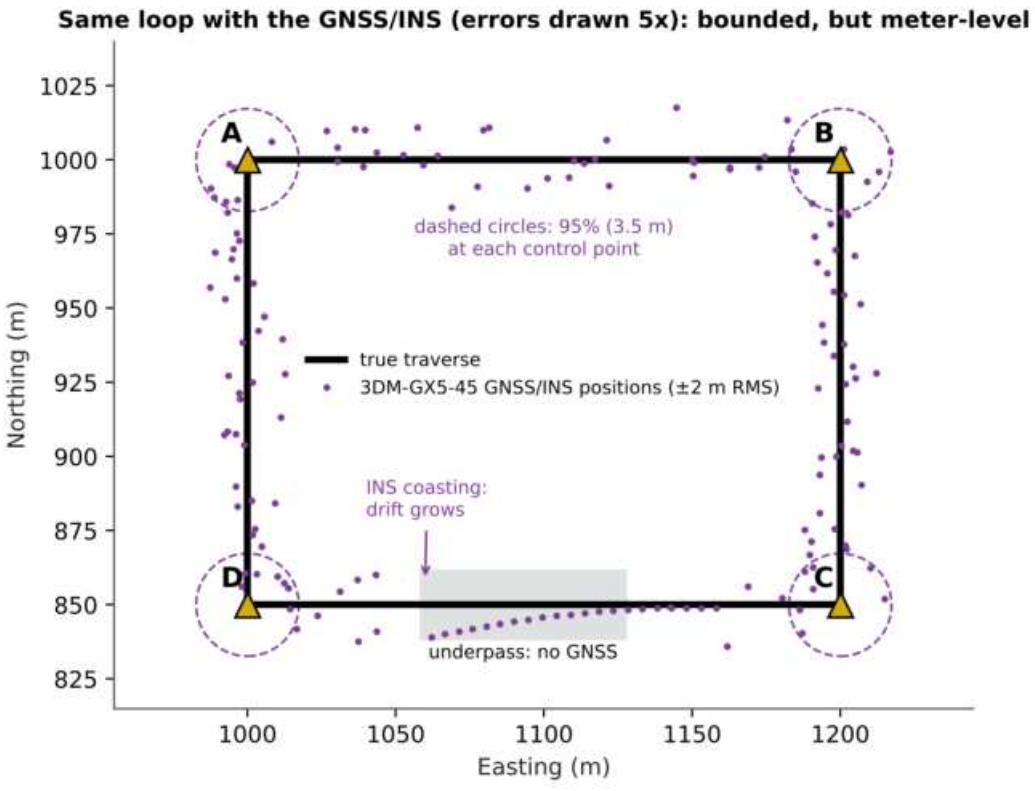


Figure 16.3: The GNSS/INS keeps every position within a few meters; in the underpass it coasts on the IMU and drift begins to grow. Errors drawn 5 times actual size.

16.2 Stepping up: the 3DM-GQ7 dual-antenna RTK GNSS/INS

SIMPLE APPROACH

Now give the second friend a much better step-counter, a pair of ears far apart that hear exactly which way the street runs, and phone messages that say "you are here, give or take two centimeters" instead of two meters. That friend can walk the block and point to the mailbox almost perfectly — as long as the phone has a signal.

Why two antennas give heading

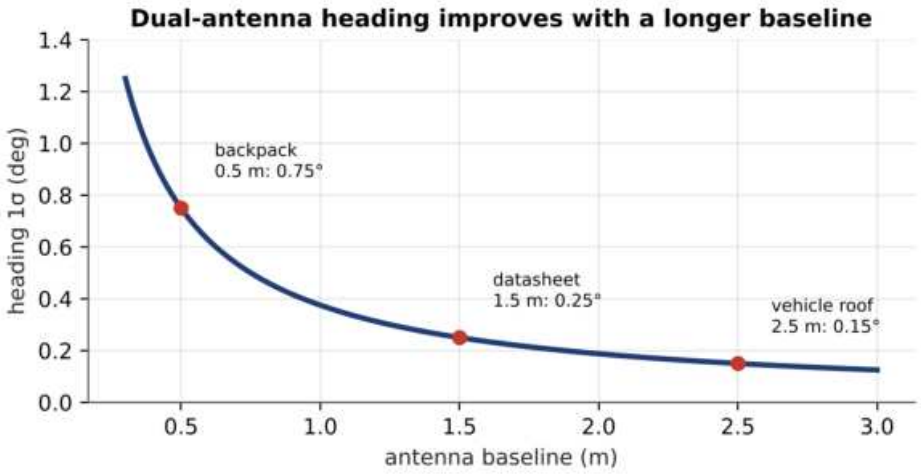


Figure 16.4: The same relative-position precision gives better heading over a longer baseline.

Real-time corrections in Florida: the FDOT FPRN

SIMPLE APPROACH

RTK is like having a friend standing on a spot whose location is known exactly. Your friend radios you, "The satellites are fooling me by this much right now." You subtract the same mistake from your own reading and suddenly you know where you are to within a couple of centimeters. In Florida, the state runs over a hundred of these "friends" for free.

SURVEYOR'S FIELD TIP

Standards of practice for network RTK. Before and after each session, *check in on a known control point* and record the difference; log the mountpoint, solution type (VRS/iMAX/single), fix status, PDOP and the number of satellites; record the network datum and epoch (NAD 83(2011) epoch 2010.0) and the geoid model used; and keep independent checkpoints out of the adjustment (Module 13). This is the documentation a reviewer will expect when the accuracy statement cites RTK positioning under Florida's Chapter 5J-17 and FDOT practice.

How coasting errors grow

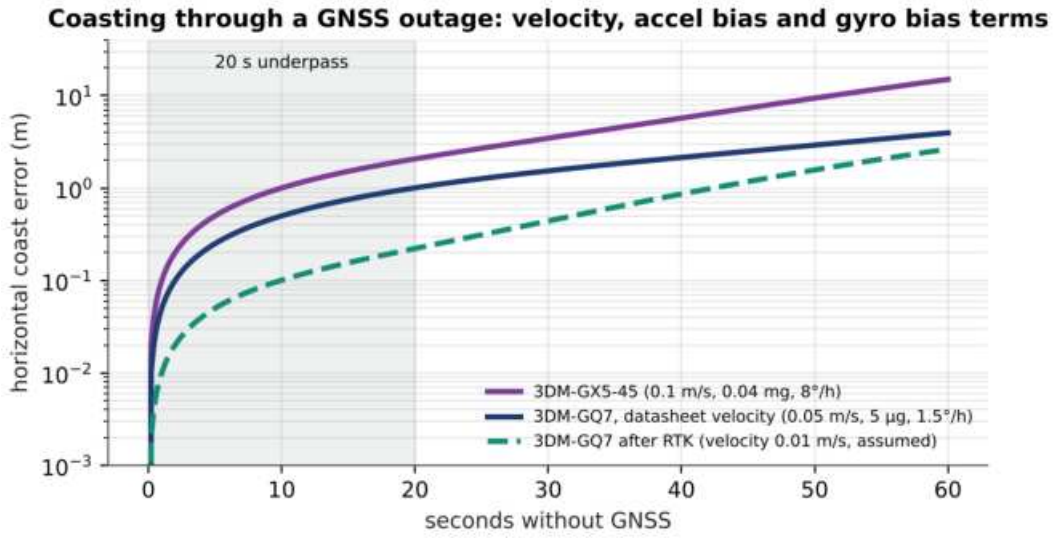


Figure 16.5: Coast error versus outage length. Velocity dominates first; the gyro's cubic term dominates later.

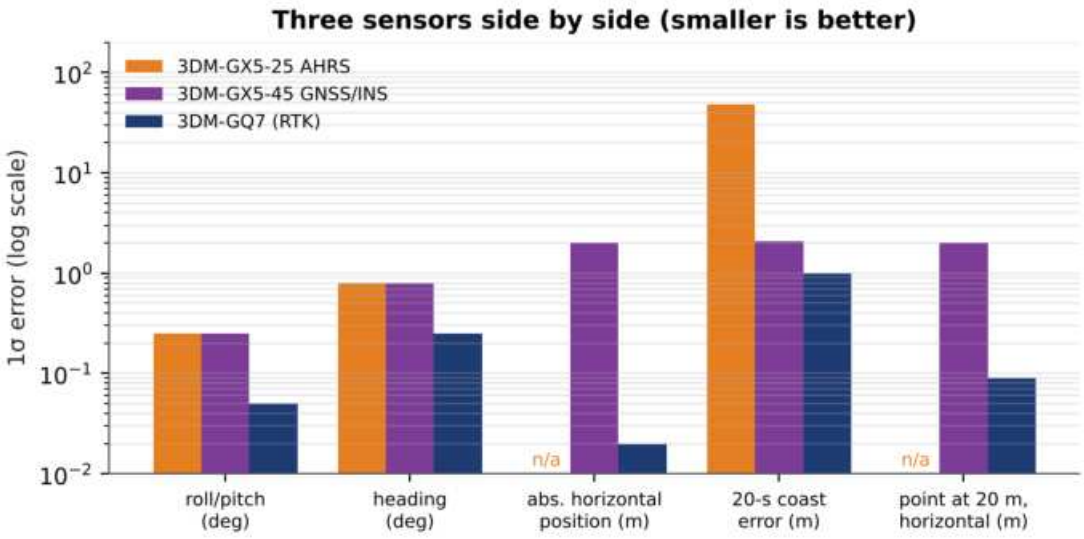


Figure 16.6: The three sensors side by side. For the AHRS, the "20-s coast" bar is free-inertial drift with an uncalibrated accelerometer bias; it has no absolute position.

16.3 The right job for each sensor: inside SLAM

SURVEYOR'S FIELD TIP

Magnetometer hygiene. Perform the hard- and soft-iron calibration with the sensor mounted on the final rig (the rig's own steel and motors are the biggest disturbance), keep the unit as far as practical from motors and batteries, and treat magnetic heading as suspect near vehicles, rebar, steel bridges and buried utilities — Run 2's 2° bridge error is a conservative example.

MODULE 17

Survey-Grade Case Study: RIEGL VUX-240[24]

SIMPLE APPROACH

This scanner is like a super-fast, super-careful version of the ones we met before. It fires up to 2.4 million laser flashes a second from an airplane. To know exactly where each dot lands 500 feet below, we have to add up all the little mistakes: the laser's own wobble, the tiny tilt and turn errors of the motion sensor, and where the GPS thinks the airplane is. The surprise: with a scanner this good, the *airplane's position* is the biggest mistake, not the laser.

17.1 The project

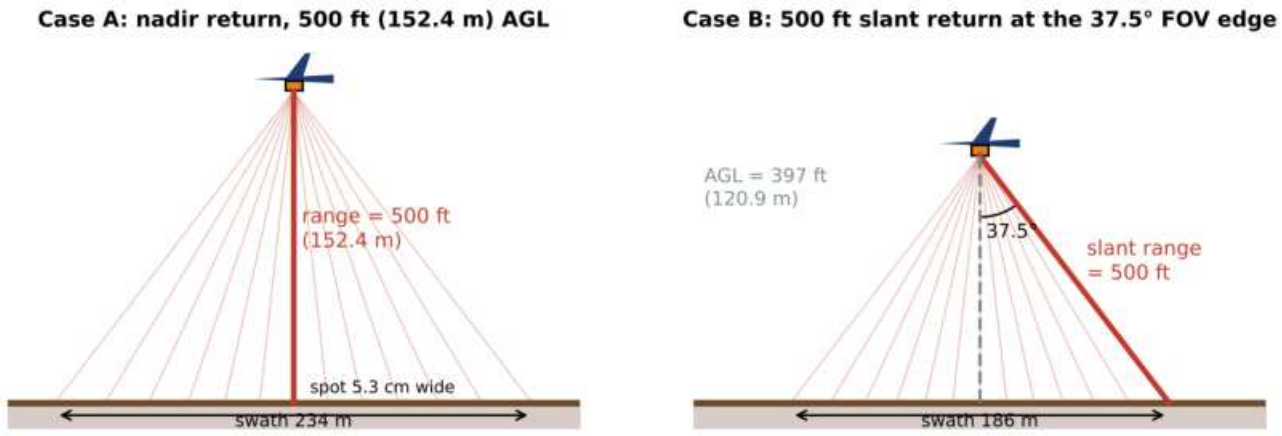


Figure 17.1: The two returns we analyze: Case A straight down from 500 ft AGL; Case B a 500 ft slant return at the edge of the 75° field of view (aircraft at 397 ft AGL).

17.2 Laser errors to consider

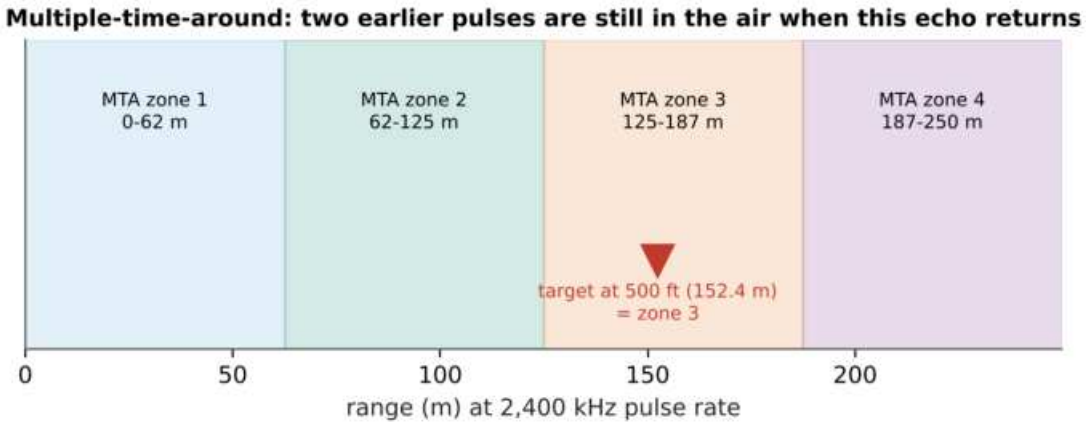


Figure 17.2: At 2.4 MHz the 500 ft echo belongs to the third pulse back.

17.3 Survey-grade checklist for this system

BIG IDEA

At 500 ft, a survey-grade laser is no longer the weak link: the RIEGL's own range error (2.5 cm) is comparable to the GNSS error, and the IMU's attitude error is 1–7 cm depending on the option. Survey-grade results come from the *whole system* — trajectory, calibration, control and checkpoints — not from the laser alone.

MODULE 18

Processing, QA/QC and Deliverables

18.1 From raw data to a clean cloud

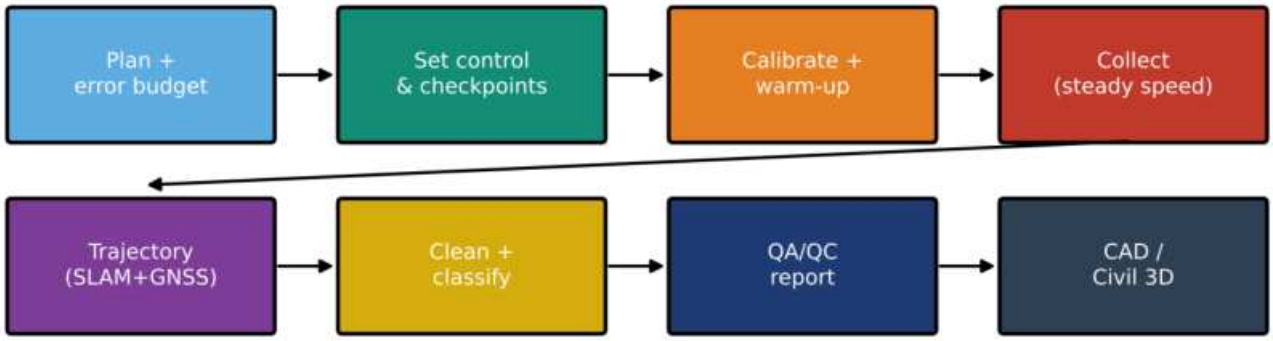


Figure 18.1: Field-to-office workflow.

18.2 Removing outliers

SIMPLE APPROACH

Imagine a crowd of people standing close together in a park — except one person way out by the fence. You can tell that person is "alone" because their nearest neighbors are far away. Outlier removal looks for dots whose nearest neighbors are far away, and throws them out: they are usually dust, rain, birds, or reflections.

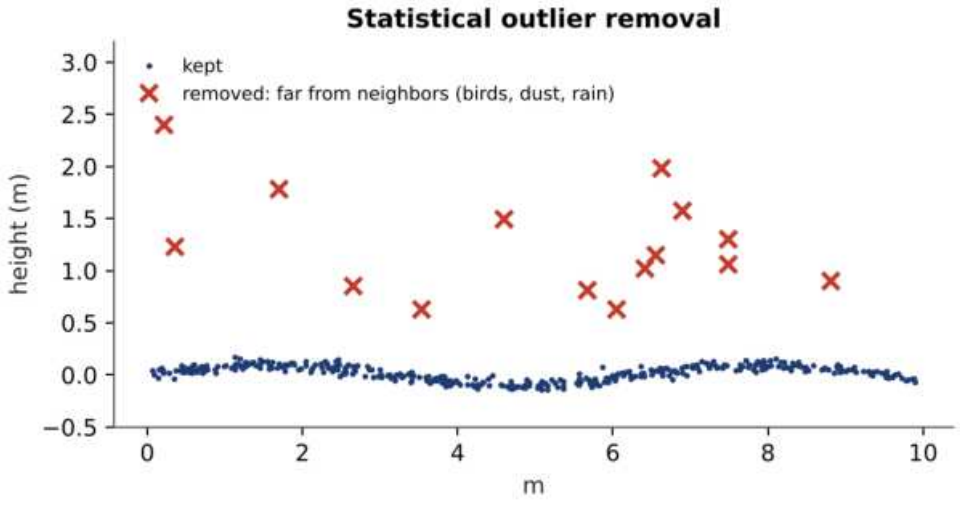


Figure 18.2: Statistical outlier removal keeps points whose neighbors are close.

18.3 Thinning: voxel down-sampling

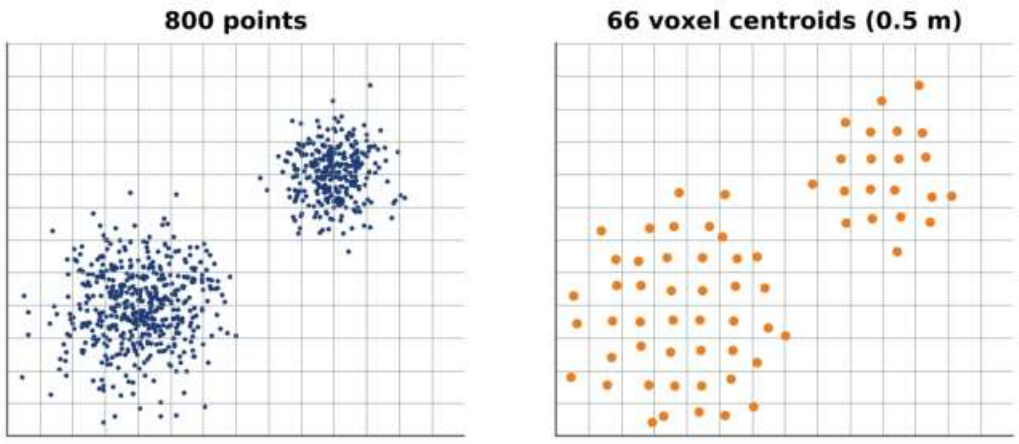


Figure 18.3: Each 0.5 m cube keeps one representative point (the centroid of its points).

WARNING — voxel size trades accuracy for speed

Making your voxel (the 3-D box) **too large reduces accuracy** while it improves performance. Every point inside a box is replaced by one average point, so a large box:

- rounds off edges, curb lips, pipe walls and other fine features — a 0.5 m voxel cannot show a 15 cm curb reveal;
- blends points from different surfaces (a wall and the ground beside it) into one point that lies on neither;
- thins the features that scan matching relies on, which can weaken SLAM and let the trajectory drift (Modules 11–12).

A small voxel keeps detail but leaves more points, so processing, matching and file sizes all cost more. **There is no single correct size: you have to balance the cost with trial and error.** Start with a voxel well below the smallest feature you must map (for example, one-third of it), process a test section, compare the result against checkpoints and the full-resolution cloud, then enlarge the voxel only as far as accuracy still meets the specification. Record the voxel size you used in the processing report.

18.4 Classification and the bare-earth surface

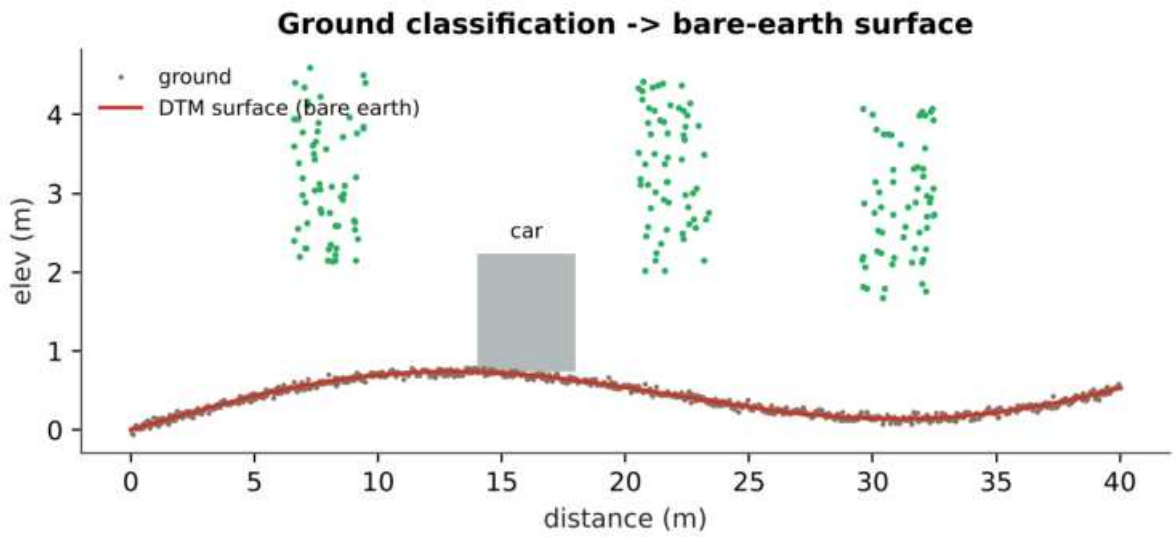


Figure 18.4: Classifying ground points and building a bare-earth surface (DTM).

18.5 Volumes

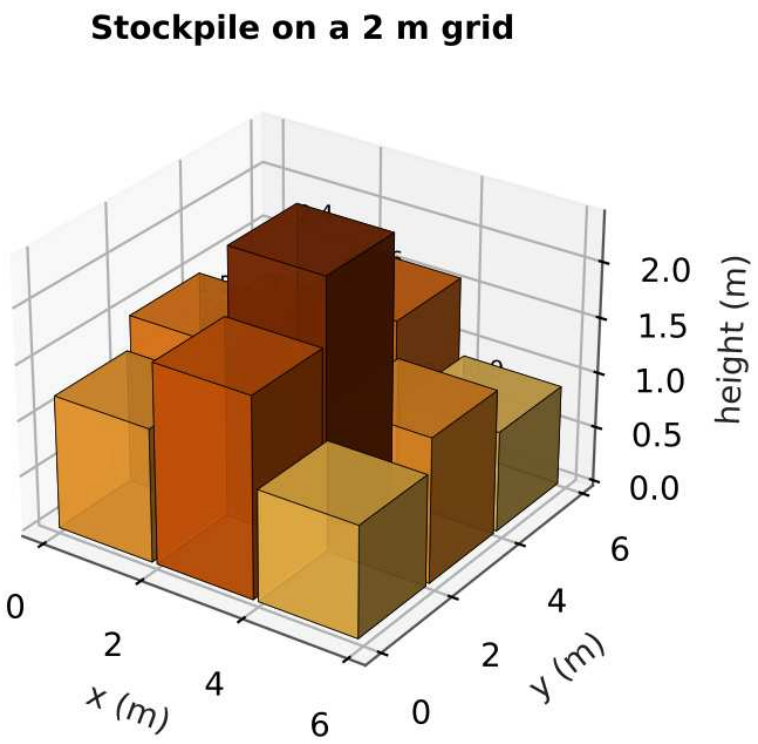


Figure 18.5: A stockpile gridded at 2 m: each column is height × cell area.

18.6 Cloud-to-cloud QA

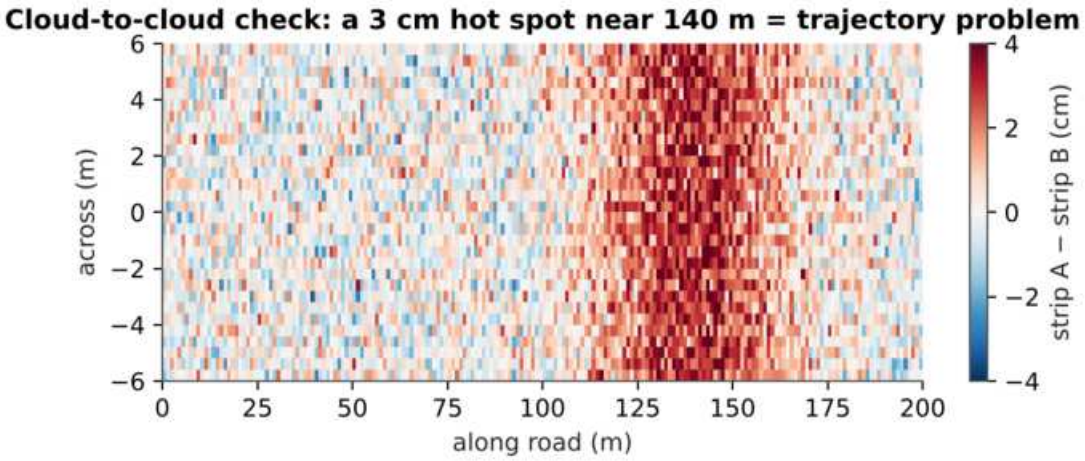


Figure 18.6: Differencing two overlapping passes reveals trajectory problems that checkpoints between them could miss.

18.7 Deliverables and CAD

SURVEYOR'S FIELD TIP

Civil 3D handles point clouds through ReCap (RCP/RCS): index the LAS/LAZ in ReCap, attach to the drawing, create a surface from the ground class, then add breaklines. Very large clouds should be tiled and thinned for the surface while the full cloud is kept for reference. Match deliverable size to what the client's hardware and software can actually open.

MODULE 19

Florida Standards of Practice, FDOT and the New Datums

REGULATORY REVIEW REQUIRED BEFORE TEACHING

This module was compiled from the rule and agency texts listed at the end of the module (retrieved October 2026). It is **not legal advice** and has **not yet been reviewed by a second Florida PSM**. Before the seminar, a reviewing PSM should check every citation against the current official text and sign below.

Reviewer: _____, PSM No. _____ Date: _____ Rule versions checked: _____

SIMPLE APPROACH

Every game has a rulebook. Surveyors in Florida have one too, written by the state. It says what every map must show, how accurate it must be and how you prove it, and how you sign your name to it. When the map is made with a laser scanner instead of a total station, the rulebook still applies — the scanner is just a new kind of measuring tape.

19.1 General requirements applied to mobile LiDAR (5J-17.051)

SURVEYOR'S FIELD TIP

5J-17.052 still governs boundaries. A LiDAR topographic survey does not satisfy boundary-survey requirements (monuments, measured dimensions, water-boundary descriptions and so on in 5J-17.052). If a LiDAR product is part of a boundary survey, the boundary items must still be established and shown under 5J-17.052.

19.2 FDOT Terrestrial Mobile LiDAR (TML) requirements

SURVEYOR'S FIELD TIP

District practice example (FDOT District 5 QA/QC plan). "Imagery" covers aerial photogrammetry *and* TML. Coordinate with the FDOT Central Office for a Mobile Survey Tracking System (MSTS) number; submit all raw, interim and final imagery data to the Central Office with a letter of transmittal; deliver the derived DGN/TIN files to the District; and note in the Survey and Map Report that the Central Office has received a separate certified report covering the imagery data. Check your own District's plan.

19.3 The new national datums (NSRS modernization)

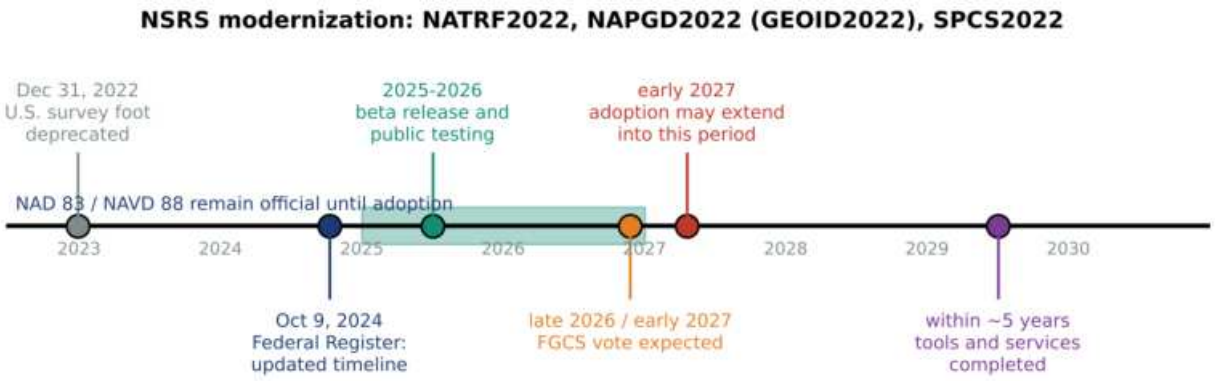


Figure 19.1: NSRS modernization timeline as published by NGS and the Federal Register. Dates after 2026 are NGS expectations, not fixed.

SURVEYOR'S FIELD TIP

Future-proof today's LiDAR. Record datum, realization, epoch, geoid model and foot in the report, the LAS header (WKT) and the metadata; keep ellipsoid heights and the trajectory so the project can be transformed to NATRF2022/NAPGD2022 later; and expect clients to ask for re-delivery in the new datums once they are adopted.

MODULE 20

Professional Practice for LiDAR Surveys

SIMPLE APPROACH

Having a great camera does not make you a professional photographer. A professional also makes sure the right person is in charge, writes down what was promised, works safely, keeps the originals, and signs only what they have checked. The same is true for a laser scanner.

MODULE 21

Capstone and Sample Project Dataset

SIMPLE APPROACH

This is the "big game" at the end of the season. You use every skill you practiced: measuring with light, turning angles into dots, keeping track of the car, fixing mistakes with landmarks, and checking your work with secret checkpoints. If every step is done carefully, the final map can be trusted to build a real road.

21.1 Collecting and processing

SURVEYOR'S FIELD TIP

Sample accuracy statement. "This dataset was tested for vertical accuracy at 30 independent checkpoints on hard surfaces and found to have RMSE_z = 2.1 cm, including checkpoint survey uncertainty, and RMSE_H = 2.8 cm horizontally, in accordance with the ASPRS Positional Accuracy Standards, Edition 2. Horizontal datum: [NAD83(2011) State Plane Florida East, US ft]; vertical datum: [NAVD88, GEOID18]." Fill in the datums your project actually used, and comply with the content requirements of the governing standards of practice (in Florida, Ch. 5J-17 F.A.C.).

21.2 Sample project dataset: trajectory, checkpoints and accuracy report

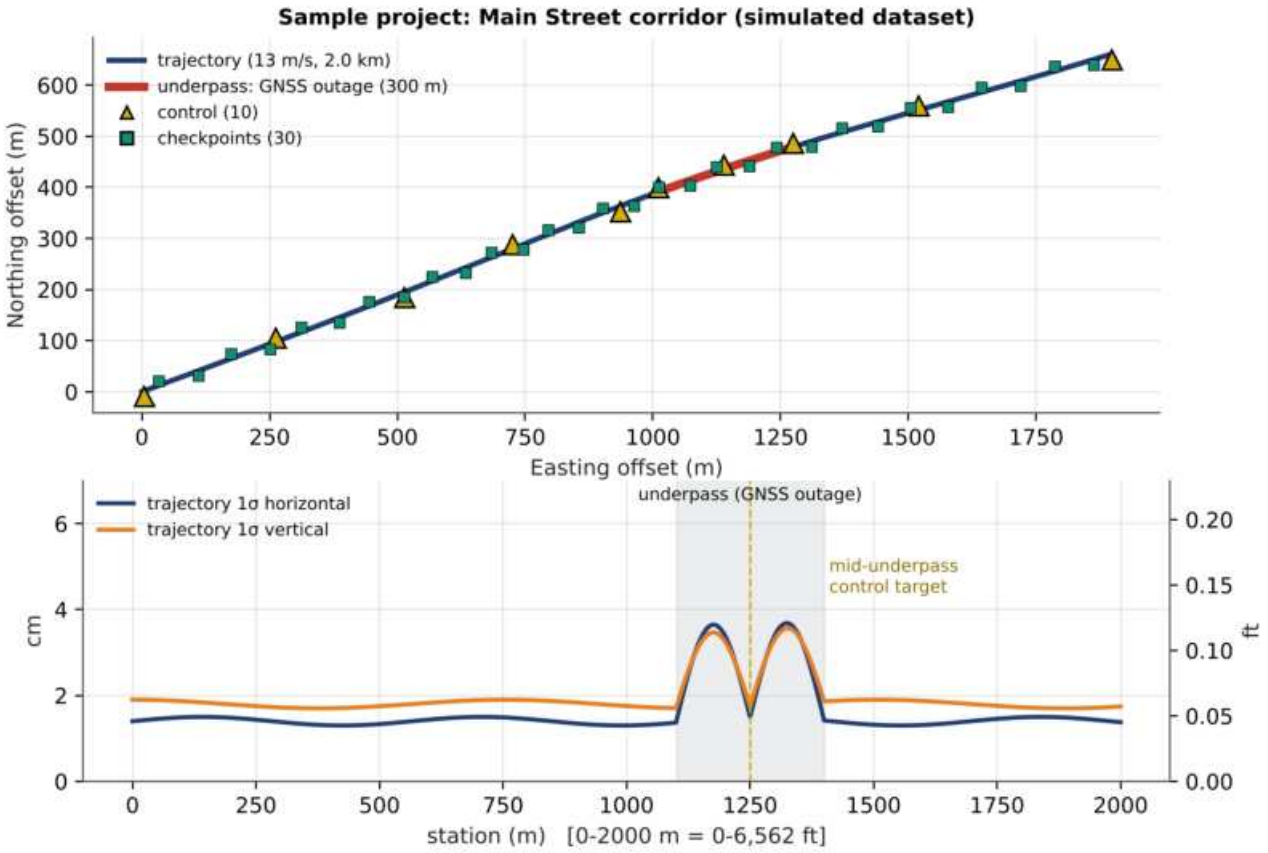


Figure 21.1: Top: the 2.0 km (6,562 ft) trajectory with 10 control points and 30 checkpoints; the red stretch is the GNSS outage under the underpass. Bottom: the trajectory's own 1σ estimates rise in the underpass and are pulled back down by the mid-underpass control target.

BIG IDEA

Read the trajectory before reading the point cloud. The bottom plot shows exactly where accuracy is strong (open sky, 1.4 cm / 0.05 ft) and where it is weak (underpass, up to 3.6 cm / 0.12 ft) — which is where extra checkpoints belong.

Checkpoint table

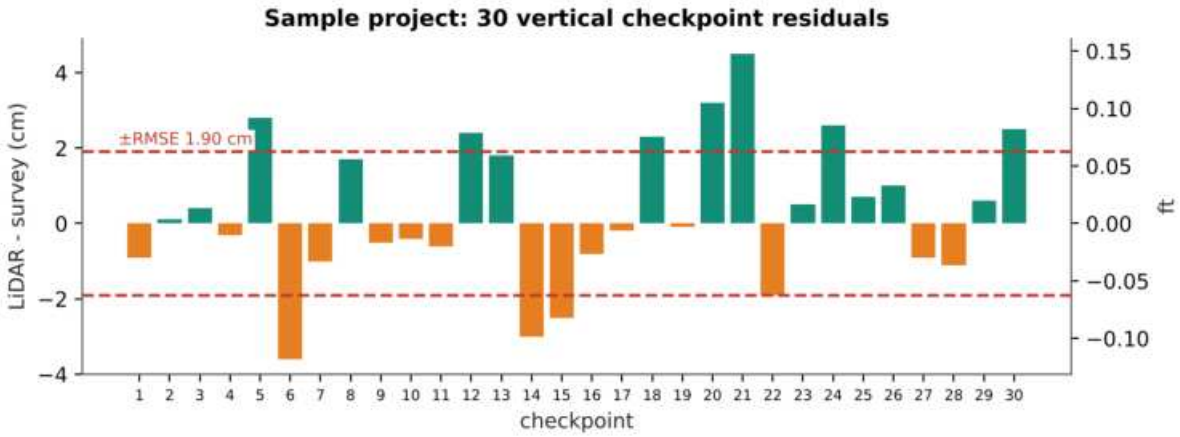


Figure 21.2: Vertical residuals at the 30 checkpoints. No residual exceeds 3×RMSE (5.7 cm), so none is flagged as a blunder.

21.3 One pulse, one more time: the master checklist

BIG IDEA

Measure carefully, control what you can, check what you must, and document what you did. That is how survey-grade work is produced with any instrument — including a laser that never stands still.